

Constant-Lagrangian inflow pipeline applied to SPARC (175 galaxies)

Single-L and two-L models of de Haas (SPARC H_z paper, Eqs. 17–18 and 24–26), $H_z = 2.2 \times 10^{-18} \text{ s}^{-1}$ fixed; fits on $v^2(r)$ with $\sigma_{v^2} = 2V_{\text{obs}}\delta V_{\text{obs}}$; data Lelli, McGaugh & Schombert (2016).

Pipeline

1. Single-L fit (R, M) to every galaxy (14 starting radii, bounded least squares).
2. Wave screen on the weighted residuals $\text{WR}_i = (v_{\text{obs}}^2 - v_{\text{mod}}^2)/\sigma_{v^2}$: one-sided Wald–Wolfowitz runs test (p_{runs} , too few sign runs) and the best agreement A_{+-+} (A_{-+-}) of the residual signs with a three-segment template (each segment ≥ 2 points). Categories: **A** adequate ($p_{\text{runs}} \geq 0.05$, $\chi_\nu^2 \leq 1$); **B** $+-+$ wave ($p_{\text{runs}} < 0.05$, $A_{+-+} \geq 0.8$, $A_{+-+} > A_{-+-}$); **C** $-+-$ wave (same with signs reversed); **D** structured, other ($p_{\text{runs}} < 0.05$); **E** poor, unstructured ($p_{\text{runs}} \geq 0.05$, $\chi_\nu^2 > 1$).
3. Category B: two-L fit seeded at $R_2 \simeq r_{\times 1}/0.8$ ($r_{\times 1} = \text{first} + \rightarrow - \text{crossover}$), cross-checked with a global grid search; also two-L + Φ_{BH} .
4. Accept the two-L description when $\Delta\text{BIC} = \text{BIC}_{2\text{L}} - \text{BIC}_{1\text{L}} < -6$.
5. Accepted fits with $n \leq 13$ are marked tentative.

Post-hoc flags on accepted fits (not part of the acceptance rule): **N** = not a nested configuration ($M_2 < M_1$, or v^2 drops by more than 40% at R_2); **M** = still inadequate after two-L ($\chi_\nu^2 > 2$), i.e. more regions (virial windows, a third Lagrangian) needed.

Table 1: Pipeline outcome.

Category after single-L fit	N
A: single-L adequate	54
B: $+-+$ wave (double-L candidate)	45
C: $-+-$ wave (declining / virial)	41
D: structured, other	18
E: poor, unstructured	17
B: two-L accepted ($\Delta\text{BIC} < -6$)	23
of which tentative ($n \leq 13$)	5
of which flagged N (not nested)	5
of which flagged M (still $\chi_\nu^2 > 2$)	8
B: single-L kept ($\Delta\text{BIC} \geq -6$)	22
2018 RMWRSS double-fit galaxies recovered as accepted two-L	8 / 13
Seeded two-L start reached the global-grid optimum	31 / 45

Table 2: Single-L fit and wave screen for all 175 SPARC galaxies, sorted by category and χ^2_ν . R in kpc, M in $10^9 M_\odot$; † = 2018 RMWRSS double-fit galaxy.

Galaxy	Type	Q	n	R	M	χ^2_ν	RMS _{rel}	P_{runs}	A_{++}	A_{+-}	Cat	ΔBIC	Outcome
UGC09992	Im	2	5	0.431 ± 0.4	0.0414 ± 0.047	0.00	0.018	1.000	—	—	A	—	
UGC06628	Sm	2	7	1.05 ± 0.41	0.165 ± 0.088	0.00	0.025	0.113	0.29	0.86	A	—	
F561–1	Sm	3	6	2.22 ± 0.74	0.524 ± 0.25	0.01	0.021	0.500	0.50	0.50	A	—	
UGC05999	Im	2	5	4.83 ± 0.59	4.62 ± 0.9	0.02	0.013	1.000	—	—	A	—	
NGC3949	Sbc	2	7	1.34 ± 0.17	3.27 ± 0.6	0.03	0.019	0.358	0.86	0.29	A	—	
F563–V2	Im	1	10	1.83 ± 0.24	2.33 ± 0.53	0.03	0.048	0.163	0.50	0.90	A	—	
F574–2	Sm	3	5	5.57 ± 3.1	1.01 ± 1.1	0.03	0.143	1.000	—	—	A	—	
UGC05750	Sdm	1	11	4.78 ± 0.58	2.8 ± 0.79	0.04	0.065	0.225	0.82	0.64	A	—	
F571–V1	Sd	2	7	3.54 ± 0.76	2.38 ± 0.74	0.04	0.060	0.358	0.71	0.57	A	—	
CamB	Im	2	9	1.37 ± 0.28	0.0865 ± 0.048	0.04	0.044	0.500	0.67	0.78	A	—	
F565–V2	Im	2	7	3.56 ± 0.61	2.5 ± 0.74	0.04	0.085	0.113	0.86	0.29	A	—	
UGC08837	Im	2	8	2.77 ± 0.31	0.915 ± 0.26	0.05	0.041	0.223	0.88	0.50	A	—	
UGC07577	Im	2	9	1.23 ± 0.53	0.0556 ± 0.061	0.06	0.272	0.051	0.89	0.67	A	—	
D512–2	Im	2	4	0.997 ± 0.16	0.132 ± 0.041	0.07	0.055	1.000	—	—	A	—	
UGC10310	Sm	1	7	1.58 ± 0.25	0.758 ± 0.18	0.07	0.049	0.560	0.43	0.71	A	—	
UGC02023	Im	2	5	2.97 ± 1.4	1.66 ± 2	0.08	0.172	1.000	—	—	A	—	
NGC4068	Im	2	6	1.44 ± 0.21	0.297 ± 0.1	0.08	0.077	0.819	0.50	0.50	A	—	
UGC07261	Sdm	2	7	0.922 ± 0.14	0.431 ± 0.09	0.09	0.041	0.113	0.86	0.43	A	—	
UGC06930	Sd	1	10	2.19 ± 0.34	2.19 ± 0.43	0.10	0.044	0.103	0.70	0.90	A	—	
UGC07559	Im	2	7	1.05 ± 0.19	0.115 ± 0.04	0.11	0.152	0.181	1.00	0.29	A	—	
NGC4085	Sc	2	7	1.9 ± 0.15	3.49 ± 0.46	0.13	0.038	0.560	0.43	0.71	A	—	
F563–V1	Im	3	6	1.98 ± 0.77	0.159 ± 0.096	0.14	0.110	0.240	0.00	1.00	A	—	
F567–2	Sm	3	5	2.17 ± 0.67	0.527 ± 0.25	0.15	0.089	1.000	—	—	A	—	
NGC3877	Sc	2	13	2.27 ± 0.17	6.15 ± 0.61	0.15	0.062	0.076	0.46	0.92	A	—	
D564–8	Im	2	6	1.07 ± 0.13	0.0647 ± 0.015	0.15	0.092	0.181	0.83	0.17	A	—	
UGC07608	Im	1	8	1.8 ± 0.31	0.809 ± 0.26	0.16	0.092	0.223	0.88	0.50	A	—	
UGC04483	Im	2	8	0.344 ± 0.049	0.0189 ± 0.0042	0.17	0.188	0.582	0.88	0.38	A	—	
UGC11557	Sdm	2	12	3.2 ± 0.42	2.26 ± 0.52	0.18	0.308	0.179	0.58	0.83	A	—	
UGC05918	Im	2	8	1.08 ± 0.18	0.18 ± 0.047	0.19	0.129	0.063	0.88	0.50	A	—	
UGC05005	Im	1	11	5.97 ± 0.72	5.36 ± 1.2	0.19	0.278	0.074	0.91	0.55	A	—	
UGC07866	Im	2	7	0.671 ± 0.16	0.0598 ± 0.024	0.21	0.170	0.113	1.00	0.14	A	—	
F583–1	Sm	1	25	2.92 ± 0.19	1.78 ± 0.22	0.21	0.250	0.059	0.88	0.64	A	—	
UGC06399	Sm	1	9	2.28 ± 0.27	1.64 ± 0.3	0.22	0.111	0.051	1.00	0.56	A	—	
UGCA281	BCD	3	7	0.335 ± 0.037	0.0281 ± 0.0055	0.29	0.186	0.736	0.86	0.43	A	—	
UGC04499	Sdm	1	9	1.89 ± 0.14	0.935 ± 0.1	0.35	0.100	0.051	1.00	0.56	A	—	
UGC06973	Sab	3	9	0.157 ± 0.2	0.373 ± 0.49	0.36	0.036	0.051	1.00	0.56	A	—	
NGC3917	Scd	1	17	3.09 ± 0.13	5.51 ± 0.35	0.40	0.040	0.165	0.59	0.88	A	—	
UGC04325	Sm	1	8	1.01 ± 0.074	0.778 ± 0.08	0.41	0.049	0.500	0.50	0.75	A	—	
UGC07151	Scd	1	11	1.02 ± 0.061	0.483 ± 0.043	0.45	0.070	0.225	0.91	0.55	A	—	
UGC07603	Sd	1	12	0.871 ± 0.067	0.317 ± 0.036	0.46	0.126	0.113	0.75	0.75	A	—	
DDO170	Im	2	8	2.92 ± 0.15	1.02 ± 0.075	0.50	0.076	0.937	0.62	0.62	A	—	
NGC3953	Sbc	1	8	2.12 ± 0.3	9.34 ± 1.6	0.52	0.048	0.075	0.38	1.00	A	—	
F583–4†	Sc	1	12	1.36 ± 0.17	0.487 ± 0.085	0.56	0.239	0.126	0.92	0.50	A	—	
NGC6789	BCD	2	4	0.41 ± 0.12	0.159 ± 0.11	0.58	0.258	1.000	—	—	A	—	
UGC07690	Im	2	7	0.349 ± 0.11	0.101 ± 0.039	0.66	0.103	0.113	0.43	0.86	A	—	
UGC02259	Sdm	2	8	0.98 ± 0.073	0.625 ± 0.057	0.66	0.052	0.582	0.88	0.38	A	—	
UGC06818	Sm	2	8	3.77 ± 0.45	2.58 ± 0.7	0.68	0.274	0.223	0.75	0.75	A	—	
UGC07089	Sdm	2	12	2.83 ± 0.27	1.49 ± 0.25	0.70	0.199	0.113	0.92	0.58	A	—	
UGC07232	Im	2	4	0.504 ± 0.11	0.12 ± 0.057	0.71	0.186	1.000	—	—	A	—	
UGC06923	Im	2	6	1.82 ± 0.27	1.25 ± 0.3	0.71	0.226	0.500	0.17	0.83	A	—	
UGC06667	Scd	1	9	2.25 ± 0.15	1.58 ± 0.17	0.72	0.145	0.110	1.00	0.56	A	—	
UGC06917	Sm	1	11	2.15 ± 0.14	2.18 ± 0.21	0.74	0.086	0.175	0.82	0.55	A	—	
NGC4389	Sbc	3	6	2.83 ± 0.39	3.63 ± 1.1	0.75	0.176	0.181	0.83	0.17	A	—	
UGC05414	Im	1	6	1.57 ± 0.14	0.567 ± 0.088	0.93	0.137	0.181	0.83	0.17	A	—	
F568–1	Sc	1	12	2.67 ± 0.32	4.29 ± 0.81	0.05	0.094	0.008	1.00	0.50	B	4.6	single-L kept
F579–V1	Sc	1	14	0.983 ± 0.23	1 ± 0.29	0.10	0.108	0.003	0.93	0.79	B	4.5	single-L kept
UGC08286	Scd	1	17	1.12 ± 0.061	0.667 ± 0.046	0.15	0.028	0.019	0.94	0.71	B	5.5	single-L kept
F574–1	Sd	1	14	2.31 ± 0.16	1.99 ± 0.2	0.16	0.109	0.003	1.00	0.71	B	3.7	single-L kept
KK98–251	Im	2	15	1.55 ± 0.11	0.228 ± 0.038	0.18	0.066	0.013	0.93	0.60	B	4.6	single-L kept
F568–3	Sd	1	18	3.65 ± 0.24	3.86 ± 0.54	0.19	0.092	0.014	0.94	0.72	B	4.0	single-L kept
NGC5005	Sbc	1	18	0.397 ± 0.13	2.11 ± 0.77	0.20	0.087	0.001	1.00	0.78	B	4.8	single-L kept
UGCA444	Im	2	36	0.759 ± 0.074	0.0805 ± 0.014	0.26	0.223	7.6e-06	0.97	0.64	B	1.2	single-L kept
UGC12632	Sm	1	15	2.18 ± 0.15	1.02 ± 0.1	0.28	0.179	0.002	1.00	0.60	B	3.7	single-L kept
NGC0100	Scd	1	21	2.27 ± 0.14	1.56 ± 0.16	0.37	0.146	0.002	0.90	0.57	B	0.2	single-L kept
NGC2976	Sc	2	27	1.24 ± 0.056	1.21 ± 0.14	0.47	0.155	0.013	0.85	0.74	B	1.1	single-L kept
NGC4214	Im	2	14	0.188 ± 0.041	0.09 ± 0.022	0.53	0.122	0.010	0.93	0.79	B	-0.6	single-L kept

Galaxy	Type	Q	n	R	M	χ^2_ν	RMS _{rel}	<i>p</i> _{pruns}	A ₊₊	A ₊₋	Cat	Δ BIC	Outcome
UGC06446 [†]	Sd	1	17	1.22 ± 0.14	0.679 ± 0.098	0.67	0.147	0.004	0.94	0.82	B	-2.9	single-L kept
UGC08550	Sd	1	11	0.685 ± 0.057	0.179 ± 0.019	0.68	0.061	0.013	1.00	0.64	B	2.9	single-L kept
NGC3109 [†]	Sm	1	25	2.26 ± 0.075	0.942 ± 0.061	0.78	0.127	2.6e-04	0.96	0.68	B	-5.9	single-L kept
UGC05829 [†]	Im	2	11	2.97 ± 0.38	1.32 ± 0.32	0.83	0.286	0.013	1.00	0.64	B	-1.8	single-L kept
NGC0055	Sm	2	21	3.14 ± 0.1	2.12 ± 0.12	1.04	0.135	1.3e-04	1.00	0.76	B	-8.1	two-L
UGC07524	Sm	1	31	2.78 ± 0.1	1.67 ± 0.1	1.07	0.249	6.7e-05	0.94	0.71	B	-16.6	two-L
ESO079-G014	Sbc	1	15	5.09 ± 0.18	15.9 ± 0.84	1.12	0.345	0.009	0.93	0.67	B	-1.2	single-L kept
UGC07125	Sm	1	13	2.94 ± 0.24	1.07 ± 0.12	1.15	0.150	0.007	1.00	0.69	B	-5.4	single-L kept
UGC07399	Sdm	1	10	0.852 ± 0.059	0.7 ± 0.066	1.29	0.098	0.025	0.90	0.70	B	-3.1	single-L kept
D631-7 [†]	Im	1	16	2.44 ± 0.13	0.79 ± 0.08	1.33	0.184	0.002	1.00	0.56	B	-10.4	two-L
NGC3972 [†]	Sbc	1	10	2.14 ± 0.16	3.28 ± 0.38	1.42	0.138	0.022	1.00	0.50	B	-5.5	single-L kept
UGC06614	Sa	1	13	1.69 ± 0.87	4.6 ± 2.6	1.62	0.158	0.005	0.92	0.77	B	-9.8	two-L (tent.) [N]
UGC07323	Sdm	1	10	2.62 ± 0.2	2.03 ± 0.31	1.68	0.284	0.036	1.00	0.50	B	-5.6	single-L kept
NGC4010	Sd	2	12	3.28 ± 0.22	5.19 ± 0.55	1.69	0.239	0.038	0.83	0.75	B	-2.7	single-L kept
UGC04278 [†]	Sd	1	25	3.97 ± 0.27	4.1 ± 0.66	1.97	0.381	1.4e-05	1.00	0.68	B	-31.6	two-L
UGC02885	Sc	1	19	0.137 ± 0.18	0.867 ± 1.1	2.03	0.098	0.005	0.89	0.79	B	-17.3	two-L [N]
DDO161 [†]	Im	1	31	3.84 ± 0.13	1.65 ± 0.083	2.15	0.348	9.0e-05	0.90	0.65	B	-45.1	two-L
UGC00731	Im	1	12	1.82 ± 0.11	0.813 ± 0.069	2.34	0.175	0.008	1.00	0.67	B	-17.1	two-L (tent.)
NGC7793	Sd	1	46	0.64 ± 0.033	0.481 ± 0.037	2.57	0.303	4.4e-08	0.93	0.78	B	-67.5	two-L
ESO116-G012	Sd	1	15	1.97 ± 0.08	2.18 ± 0.13	2.57	0.259	0.007	1.00	0.60	B	-11.7	two-L
UGC12732 [†]	Sm	1	16	2.57 ± 0.2	1.73 ± 0.19	2.81	0.209	9.9e-04	1.00	0.69	B	-23.0	two-L
NGC3726	Sc	2	12	3.22 ± 0.25	7.42 ± 0.76	2.98	0.099	0.038	0.83	0.75	B	-7.3	two-L (tent.) [NM]
NGC3741 [†]	Im	1	21	1.76 ± 0.093	0.387 ± 0.036	3.52	0.365	7.5e-05	1.00	0.62	B	-54.0	two-L
F571-8 [†]	Sc	1	13	3.08 ± 0.18	5.27 ± 0.49	3.79	0.347	0.005	1.00	0.62	B	-32.6	two-L (tent.)
NGC1003	Scd	1	36	2.8 ± 0.12	2.64 ± 0.14	4.91	0.148	1.0e-04	0.92	0.69	B	-71.2	two-L [M]
NGC4013	Sb	2	36	0.02 ± 0.34	0.05 ± 0.86	5.52	0.122	2.0e-07	0.92	0.83	B	-93.7	two-L [NM]
DDO154	Im	2	12	1.76 ± 0.032	0.409 ± 0.012	7.43	0.210	0.038	0.83	0.75	B	-41.6	two-L (tent.) [M]
NGC0247 [†]	Sd	2	26	1.76 ± 0.034	1.05 ± 0.035	8.58	0.243	8.7e-06	1.00	0.85	B	-189.9	two-L
UGC03580	Sa	2	47	1.57 ± 0.028	1.73 ± 0.04	9.69	0.388	1.2e-10	1.00	0.81	B	-286.9	two-L [M]
IC2574 [†]	Sm	2	34	5.44 ± 0.099	2.58 ± 0.12	13.13	0.251	6.8e-06	0.94	0.68	B	-374.1	two-L
UGC06787	Sab	2	71	0.02 ± 0.023	0.0892 ± 0.1	25.98	0.247	1.7e-13	0.87	0.73	B	-623.0	two-L [NM]
NGC5585	Sd	1	24	1.94 ± 0.038	1.39 ± 0.045	28.93	0.400	1.1e-04	0.92	0.67	B	-529.6	two-L [M]
NGC2403	Scd	1	73	1.13 ± 0.0085	1.54 ± 0.014	32.04	0.251	3.6e-13	0.96	0.78	B	-1718.6	two-L [M]
F568-V1	Sd	1	15	2.16 ± 0.32	2.61 ± 0.59	0.08	0.145	0.008	0.67	0.80	C	-	-
NGC1705	BCD	3	14	0.175 ± 0.046	0.0739 ± 0.021	0.15	0.058	0.005	0.71	1.00	C	-	-
UGC06983	Scd	1	17	1.94 ± 0.17	2.01 ± 0.24	0.32	0.056	0.014	0.65	0.88	C	-	-
UGC01230	Sm	1	11	2.48 ± 0.39	2.42 ± 0.52	0.37	0.128	0.013	0.55	1.00	C	-	-
NGC4559	Scd	1	32	2.05 ± 0.16	2.58 ± 0.24	0.44	0.155	1.6e-06	0.81	0.91	C	-	-
NGC3521	Sbc	1	41	0.408 ± 0.05	1.52 ± 0.21	0.45	0.061	2.9e-06	0.73	0.95	C	-	-
NGC2366	Im	3	26	1.55 ± 0.062	0.415 ± 0.028	0.47	0.165	3.2e-05	0.69	0.85	C	-	-
UGC08490	Sm	1	30	0.567 ± 0.047	0.292 ± 0.028	0.50	0.103	1.3e-04	0.80	0.93	C	-	-
NGC4183	Scd	1	23	1.93 ± 0.17	2 ± 0.22	0.54	0.145	6.9e-04	0.78	0.87	C	-	-
UGC04305	Im	3	22	0.892 ± 0.12	0.0947 ± 0.018	0.65	0.568	0.001	0.73	0.86	C	-	-
UGC12506	Scd	2	31	2.16 ± 0.35	10.2 ± 1.8	0.71	0.157	2.6e-06	0.81	0.94	C	-	-
NGC3769	Sb	2	12	1.24 ± 0.27	1.52 ± 0.38	0.79	0.097	0.011	0.67	1.00	C	-	-
UGC09037	Scd	2	22	4.13 ± 0.17	8.7 ± 0.54	0.86	0.091	2.7e-04	0.82	0.91	C	-	-
UGC05721	Sd	1	23	0.481 ± 0.041	0.256 ± 0.028	1.01	0.222	1.4e-04	0.74	0.87	C	-	-
NGC4217	Sb	1	19	1.86 ± 0.11	5.69 ± 0.43	1.11	0.106	2.1e-04	0.79	1.00	C	-	-
IC4202	Sbc	1	32	2.98 ± 0.057	16.1 ± 0.46	1.31	0.081	0.021	0.66	0.84	C	-	-
NGC4088	Sbc	1	12	1.99 ± 0.21	5.15 ± 0.66	1.81	0.105	0.008	0.67	1.00	C	-	-
NGC6503	Scd	1	31	0.303 ± 0.072	0.323 ± 0.078	1.92	0.136	0.004	0.74	0.84	C	-	-
NGC4157	Sb	1	17	1.78 ± 0.16	5.29 ± 0.56	2.10	0.129	9.9e-04	0.76	1.00	C	-	-
UGC05986	Sm	2	15	2.02 ± 0.067	2.59 ± 0.13	2.15	0.144	0.008	0.73	0.87	C	-	-
NGC5985	Sb	1	33	2.25 ± 0.12	16 ± 0.97	2.20	0.069	0.001	0.67	0.88	C	-	-
UGC11914	Sab	1	65	0.264 ± 0.014	1.76 ± 0.1	2.22	0.081	9.0e-14	0.88	0.97	C	-	-
UGC02916	Sab	2	43	0.0359 ± 0.04	0.126 ± 0.14	2.48	0.115	1.1e-06	0.72	0.88	C	-	-
NGC7331	Sb	1	36	0.02 ± 0.16	0.0892 ± 0.72	2.70	0.067	4.0e-04	0.86	0.92	C	-	-
NGC0891	Sb	1	18	0.02 ± 0.13	0.076 ± 0.49	3.06	0.097	0.002	0.78	0.94	C	-	-
ESO563-G021	Sbc	1	30	5.1 ± 0.078	47.5 ± 1.3	3.38	0.116	4.2e-06	0.83	0.93	C	-	-
UGC11455	Scd	1	36	4.01 ± 0.11	25.9 ± 0.95	3.97	0.155	2.0e-07	0.75	0.86	C	-	-
UGC08699	Sab	2	41	0.02 ± 0.049	0.0538 ± 0.13	4.58	0.114	1.2e-07	0.73	0.85	C	-	-
UGC05253	Sab	2	73	0.02 ± 0.015	0.0924 ± 0.071	5.54	0.093	3.0e-15	0.85	0.99	C	-	-
NGC3992	Sbc	1	9	0.02 ± 0.89	0.101 ± 4.5	5.62	0.105	0.039	0.67	0.89	C	-	-
UGC03205	Sab	1	48	0.551 ± 0.041	2.05 ± 0.16	5.93	0.179	1.2e-07	0.92	0.94	C	-	-
NGC3198	Sc	1	43	2.15 ± 0.057	4.13 ± 0.13	6.13	0.204	5.7e-09	0.74	0.88	C	-	-
NGC4100	Sbc	1	24	1.17 ± 0.22	3.2 ± 0.66	6.72	0.233	1.6e-05	0.79	1.00	C	-	-
NGC6946	Scd	1	58	0.867 ± 0.026	2.14 ± 0.072	9.71	0.226	8.4e-09	0.79	0.90	C	-	-

Galaxy	Type	Q	n	R	M	χ^2_ν	RMS _{rel}	p_{runs}	A_{+-+}	A_{-+-}	Cat	ΔBIC	Outcome
NGC5907	Sc	1	19	0.02 ± 0.29	0.0758 ± 1.1	10.88	0.070	2.1e-04	0.79	1.00	C	–	
NGC5371	Sbc	1	19	0.139 ± 0.13	0.543 ± 0.53	14.64	0.110	2.5e-04	0.58	1.00	C	–	
NGC5033	Sc	1	22	0.02 ± 0.14	0.0699 ± 0.47	15.11	0.119	5.3e-05	0.86	0.95	C	–	
NGC6674	Sb	1	15	0.02 ± 0.55	0.0924 ± 2.5	16.38	0.165	0.031	0.80	0.87	C	–	
NGC2903	Sbc	1	34	0.708 ± 0.026	2.28 ± 0.09	26.86	0.355	1.0e-07	0.79	1.00	C	–	
UGC02953	Sab	2	115	0.219 ± 0.0096	1.45 ± 0.064	68.64	0.232	1.2e-24	0.73	0.97	C	–	
NGC5055	Sbc	1	28	0.02 ± 0.036	0.0596 ± 0.11	128.68	0.313	2.9e-06	0.68	1.00	C	–	
UGC01281	Sdm	1	25	1.96 ± 0.15	0.691 ± 0.11	0.03	0.065	2.6e-04	0.80	0.80	D	–	
PGC51017	BCD	3	6	0.02 ± 0.24	0.000561 ± 0.0069	0.21	0.096	0.034	0.50	0.50	D	–	
NGC0024	Sc	1	29	0.749 ± 0.046	0.71 ± 0.057	0.24	0.085	0.024	0.79	0.76	D	–	
F563–1	Sm	1	17	2.63 ± 0.25	2.68 ± 0.4	0.24	0.166	0.012	0.76	0.76	D	–	
DDO064	Im	1	14	1.28 ± 0.17	0.358 ± 0.11	0.24	0.306	0.014	0.79	0.79	D	–	
NGC2915	BCD	2	30	1.23 ± 0.17	0.739 ± 0.13	0.26	0.123	0.001	0.80	0.80	D	–	
NGC0300	Sd	2	25	2.19 ± 0.083	1.73 ± 0.1	1.31	0.144	0.001	0.76	0.76	D	–	
UGC06786	S0	1	45	0.824 ± 0.046	3.35 ± 0.2	1.41	0.257	3.5e-04	0.78	0.76	D	–	
NGC0289	Sbc	2	28	0.02 ± 0.44	0.0464 ± 1	2.70	0.113	0.001	0.71	0.79	D	–	
NGC7814	Sab	1	18	0.02 ± 0.13	0.0741 ± 0.5	2.96	0.114	5.1e-05	0.89	0.89	D	–	
UGC03546	Sa	1	30	0.02 ± 0.13	0.0628 ± 0.41	3.40	0.175	1.7e-04	0.80	0.80	D	–	
NGC1090	Sbc	1	24	1.95 ± 0.091	4.34 ± 0.22	3.63	0.123	3.5e-04	0.88	0.88	D	–	
NGC4138	S0	2	7	0.02 ± 0.83	0.0456 ± 1.9	4.91	0.258	0.020	0.71	0.57	D	–	
UGC00128	Sdm	1	22	5.18 ± 0.14	7.63 ± 0.25	6.09	0.185	0.005	0.73	0.77	D	–	
NGC6015	Scd	2	44	1.24 ± 0.023	2.55 ± 0.062	8.66	0.140	2.6e-06	0.68	0.73	D	–	
NGC2841	Sb	1	50	0.02 ± 0.14	0.13 ± 0.93	9.31	0.127	2.4e-10	0.96	0.96	D	–	
UGC02487	S0	1	17	0.02 ± 0.57	0.182 ± 5.2	38.15	0.125	8.9e-05	0.88	0.88	D	–	
UGC09133	Sab	1	68	0.02 ± 0.047	0.0902 ± 0.21	64.01	0.222	8.6e-14	0.91	0.91	D	–	
UGC00891	Sm	2	5	2.93 ± 0.12	1.23 ± 0.082	1.07	0.108	1.000	–	–	E	–	
NGC3893	Sc	1	10	1.08 ± 0.16	3.15 ± 0.54	1.45	0.128	0.287	0.60	0.80	E	–	
UGCA442	Sm	1	8	1.63 ± 0.095	0.49 ± 0.04	1.47	0.214	0.268	0.88	0.62	E	–	
UGC02455	Im	3	8	3.43 ± 0.77	2.28 ± 1.4	1.60	0.369	0.075	0.88	0.62	E	–	
NGC6195	Sb	1	23	0.494 ± 0.092	2.39 ± 0.47	1.83	0.121	0.334	0.74	0.74	E	–	
UGC00634	Sm	2	4	4.55 ± 0.3	5.19 ± 0.46	1.86	0.035	1.000	–	–	E	–	
NGC2998	Sc	1	13	1.02 ± 0.12	3.64 ± 0.43	1.98	0.250	0.085	0.92	0.77	E	–	
NGC4051	Sbc	2	7	2.5 ± 0.4	5.82 ± 1.3	2.61	0.270	0.358	0.43	0.86	E	–	
DDO168	Im	2	10	1.67 ± 0.067	0.578 ± 0.05	2.92	0.235	0.103	0.80	0.70	E	–	
NGC2955	Sb	1	24	0.692 ± 0.024	3.78 ± 0.16	3.06	0.111	0.110	0.67	0.75	E	–	
ESO444–G084	Im	2	7	0.842 ± 0.076	0.288 ± 0.037	3.14	0.227	0.181	1.00	0.29	E	–	
UGC00191	Sm	1	9	1.25 ± 0.035	0.712 ± 0.026	3.26	0.131	0.870	0.78	0.44	E	–	
UGC05764	Im	2	10	0.822 ± 0.034	0.238 ± 0.014	3.79	0.194	0.556	0.50	0.80	E	–	
UGC11820	Sm	1	10	4.64 ± 0.15	3.05 ± 0.15	6.72	0.664	0.087	1.00	0.60	E	–	
UGC05716	Sm	2	12	2.41 ± 0.06	1.14 ± 0.043	9.28	0.249	0.056	1.00	0.58	E	–	
NGC2683	Sb	2	11	0.02 ± 0.47	0.0544 ± 1.3	10.79	0.321	0.058	0.73	0.73	E	–	
NGC0801	Sc	1	13	1.13 ± 0.044	4.42 ± 0.2	10.80	0.175	0.076	0.62	0.92	E	–	

Table 3: Accepted two-L galaxies: best-fit parameters (1σ , local covariance). Radii in kpc, masses in $10^9 M_\odot$, Φ_{BH} in $(\text{km s}^{-1})^2$; * preferred model (lowest BIC, simpler model preferred within $\Delta\text{BIC} < 2$). Jump = fractional change of v^2 across R_2 in the two-L fit.

Galaxy	Type	n	Model	R_1	M_1	R_2	M_2	Φ_{BH}	Jump / flags
NGC2403	Scd	73	Single-L	1.13 ± 0.0085	1.54 ± 0.014	–	–	–	-0.16 M
			Two-L	0.486 ± 0.011	0.346 ± 0.012	1.67 ± 0.016	2.42 ± 0.028	–	
			Two-L + Φ_{BH} *	0.694 ± 0.021	0.528 ± 0.023	1.87 ± 0.022	2.59 ± 0.033	1028 ± 90	
UGC06787	Sab	71	Single-L	0.02 ± 0.023	0.0892 ± 0.1	–	–	–	-0.45 NM
			Two-L	0.156 ± 0.03	0.732 ± 0.14	13.4 ± 0.16	104 ± 2.3	–	
			Two-L + Φ_{BH} *	0.214 ± 0.12	0.595 ± 0.16	13.4 ± 3.9	57.8 ± 2.3	24405 ± 16292	
NGC5585	Sd	24	Single-L	1.94 ± 0.038	1.39 ± 0.045	–	–	–	+0.96 M
			Two-L	0.254 ± 0.013	0.0359 ± 0.0029	2.41 ± 0.062	1.87 ± 0.075	–	
			Two-L + Φ_{BH} *	0.256 ± 0.017	0.0342 ± 0.003	2.05 ± 0.053	1.47 ± 0.057	17 ± 52	
IC2574 [†]	Sm	34	Single-L	5.44 ± 0.099	2.58 ± 0.12	–	–	–	+0.20
			Two-L	2.55 ± 0.053	0.518 ± 0.02	5.97 ± 0.053	3.16 ± 0.076	–	
			Two-L + Φ_{BH} *	2.79 ± 0.093	0.592 ± 0.033	6.26 ± 0.022	3.53 ± 0.057	23 ± 6	
UGC03580	Sa	47	Single-L	1.57 ± 0.028	1.73 ± 0.04	–	–	–	-0.34 M
			Two-L	1.01 ± 0.04	0.824 ± 0.05	4.52 ± 0.16	6.31 ± 0.3	–	
			Two-L + Φ_{BH} *	2.19 ± 0.17	1.66 ± 0.24	5.44 ± 0.21	5.97 ± 0.31	3601 ± 239	
NGC0247 [†]	Sd	26	Single-L	1.76 ± 0.034	1.05 ± 0.035	–	–	–	-0.08
			Two-L	1.59 ± 0.032	0.835 ± 0.031	5.42 ± 0.47	6.37 ± 0.96	–	
			Two-L + Φ_{BH} *	1.74 ± 0.077	0.88 ± 0.045	5.55 ± 0.47	6.37 ± 0.96	359 ± 127	
NGC4013	Sb	36	Single-L	0.02 ± 0.34	0.05 ± 0.86	–	–	–	-0.57 NM
			Two-L	1.26 ± 0.65	4.03 ± 2.3	13.5 ± 0.68	52.1 ± 4.5	–	
			Two-L + Φ_{BH} *	0.02 (unc.)	0.0267 (unc.)	15 (unc.)	20.6 (unc.)	19890 (unc.)	
NGC1003	Scd	36	Single-L	2.8 ± 0.12	2.64 ± 0.14	–	–	–	-0.19 M
			Two-L	2.12 ± 0.12	1.8 ± 0.12	20.3 ± 1.1	39.8 ± 4.5	–	
			Two-L + Φ_{BH} *	3.69 ± 0.43	2.53 ± 0.32	10.3 ± 0.58	9.92 ± 0.58	3370 ± 549	
NGC7793	Sd	46	Single-L	0.64 ± 0.033	0.481 ± 0.037	–	–	–	+0.00
			Two-L*	0.338 ± 0.036	0.134 ± 0.026	1.25 ± 0.089	1.23 ± 0.13	–	
			Two-L + Φ_{BH}	0.323 ± 0.042	0.129 ± 0.026	1.24 ± 0.09	1.23 ± 0.13	-93 ± 154	
NGC3741 [†]	Im	21	Single-L	1.76 ± 0.093	0.387 ± 0.036	–	–	–	+0.21
			Two-L*	0.485 ± 0.065	0.0397 ± 0.0081	2.33 ± 0.15	0.599 ± 0.069	–	
			Two-L + Φ_{BH}	0.652 ± 0.13	0.0519 ± 0.014	2.43 ± 0.21	0.6 ± 0.087	128 ± 65	
DDO161 [†]	Im	31	Single-L	3.84 ± 0.13	1.65 ± 0.083	–	–	–	+0.18
			Two-L*	1.26 ± 0.16	0.196 ± 0.044	4.27 ± 0.15	1.92 ± 0.1	–	
			Two-L + Φ_{BH}	1.65 ± 0.32	0.269 ± 0.084	4.37 ± 0.17	1.9 ± 0.1	156 ± 92	
DDO154 (tent.)	Im	12	Single-L	1.76 ± 0.032	0.409 ± 0.012	–	–	–	-0.09 M
			Two-L	1.2 ± 0.07	0.18 ± 0.021	1.97 ± 0.048	0.486 ± 0.019	–	
			Two-L + Φ_{BH} *	1.44 ± 0.11	0.228 ± 0.035	2.04 ± 0.057	0.481 ± 0.019	128 ± 43	
F571–8 [†] (tent.)	Sc	13	Single-L	3.08 ± 0.18	5.27 ± 0.49	–	–	–	+0.32
			Two-L*	1.14 ± 0.15	0.687 ± 0.17	4.13 ± 0.3	8.16 ± 0.99	–	
			Two-L + Φ_{BH}	1.19 ± 0.17	0.722 ± 0.18	4.15 ± 0.31	8.16 ± 0.99	99 ± 178	
UGC04278 [†]	Sd	25	Single-L	3.97 ± 0.27	4.1 ± 0.66	–	–	–	+0.76
			Two-L*	1.03 ± 0.1	0.238 ± 0.041	4.01 ± 0.19	4.08 ± 0.45	–	
			Two-L + Φ_{BH}	1.04 ± 0.15	0.24 ± 0.054	4.01 ± 0.37	4.07 ± 0.84	16 ± 56	
UGC12732 [†]	Sm	16	Single-L	2.57 ± 0.2	1.73 ± 0.19	–	–	–	-0.12
			Two-L	1.79 ± 0.17	0.978 ± 0.13	9.6 ± 1.2	12.2 ± 3	–	
			Two-L + Φ_{BH} *	2.76 ± 0.37	1.47 ± 0.25	10.6 ± 1.6	13.1 ± 4.2	955 ± 262	
UGC02885	Sc	19	Single-L	0.137 ± 0.18	0.867 ± 1.1	–	–	–	-0.69 N
			Two-L*	0.863 ± 0	9.42 ± 0	1.7 ± 0.25	11.3 ± 2.1	–	
			Two-L + Φ_{BH}	15.9 ± 20	11.6 ± 21	42.9 ± 22	94.1 ± 92	73141 ± 3447	
UGC00731 (tent.)	Im	12	Single-L	1.82 ± 0.11	0.813 ± 0.069	–	–	–	-0.10
			Two-L*	1.02 ± 0.12	0.26 ± 0.057	3.04 ± 0.35	1.62 ± 0.27	–	
			Two-L + Φ_{BH}	1.38 ± 0.67	0.343 ± 0.22	3.23 ± 0.42	1.62 ± 0.27	411 ± 452	
UGC07524	Sm	31	Single-L	2.78 ± 0.1	1.67 ± 0.1	–	–	–	+0.25
			Two-L*	1.06 ± 0.14	0.228 ± 0.052	3.06 ± 0.15	1.91 ± 0.15	–	
			Two-L + Φ_{BH}	1.48 ± 0.25	0.363 ± 0.12	3.1 ± 0.16	1.88 ± 0.15	188 ± 84	

Galaxy	Type	n	Model	R_1	M_1	R_2	M_2	Φ_{BH}	Jump / flags
ESO116-G012	Sd	15	Single-L	1.97 ± 0.08	2.18 ± 0.13	—	—	—	
			Two-L	0.927 ± 0.13	0.412 ± 0.11	2.39 ± 0.12	2.82 ± 0.21	—	+0.19
			Two-L + Φ_{BH}^*	1.61 ± 0.18	1.24 ± 0.29	2.68 ± 0.17	3.17 ± 0.29	432 ± 139	
D631-7 [†]	Im	16	Single-L	2.45 ± 0.13	0.79 ± 0.08	—	—	—	
			Two-L*	1.43 ± 0.16	0.242 ± 0.056	3.02 ± 0.2	1.15 ± 0.15	—	+0.08
			Two-L + Φ_{BH}	1.43 ± 0.19	0.243 ± 0.061	3.02 ± 0.2	1.15 ± 0.15	1 ± 33	
UGC06614 (tent.)	Sa	13	Single-L	1.69 ± 0.87	4.6 ± 2.6	—	—	—	
			Two-L*	$4.28 \pm 7e+06$	$65.7 \pm 3.2e+08$	5.41 ± 1.1	18.2 ± 4.4	—	-0.85 N
			Two-L + Φ_{BH}	$2.78 \pm 3.3e+07$	$1e-05 \pm 1.1$	$16.9 \pm 1.3e+02$	20.6 ± 43	29531 ± 127786	
NGC0055	Sm	21	Single-L	3.14 ± 0.1	2.12 ± 0.12	—	—	—	
			Two-L*	1.52 ± 0.25	0.381 ± 0.11	3.68 ± 0.21	2.72 ± 0.27	—	+0.34
			Two-L + Φ_{BH}	3.38 ± 1.8	1.89 ± 2.9	4.03 ± 0.25	2.85 ± 0.3	593 ± 205	
NGC3726 (tent.)	Sc	12	Single-L	3.22 ± 0.25	7.42 ± 0.76	—	—	—	
			Two-L*	3.57 ± 0.26	8.99 ± 0.96	13.9 ± 1.5	48.8 ± 9.9	—	-0.45 NM
			Two-L + Φ_{BH}	4.76 ± 1	9.94 ± 1.8	13.9 ± 2.5	35.8 ± 9.8	6415 ± 3580	

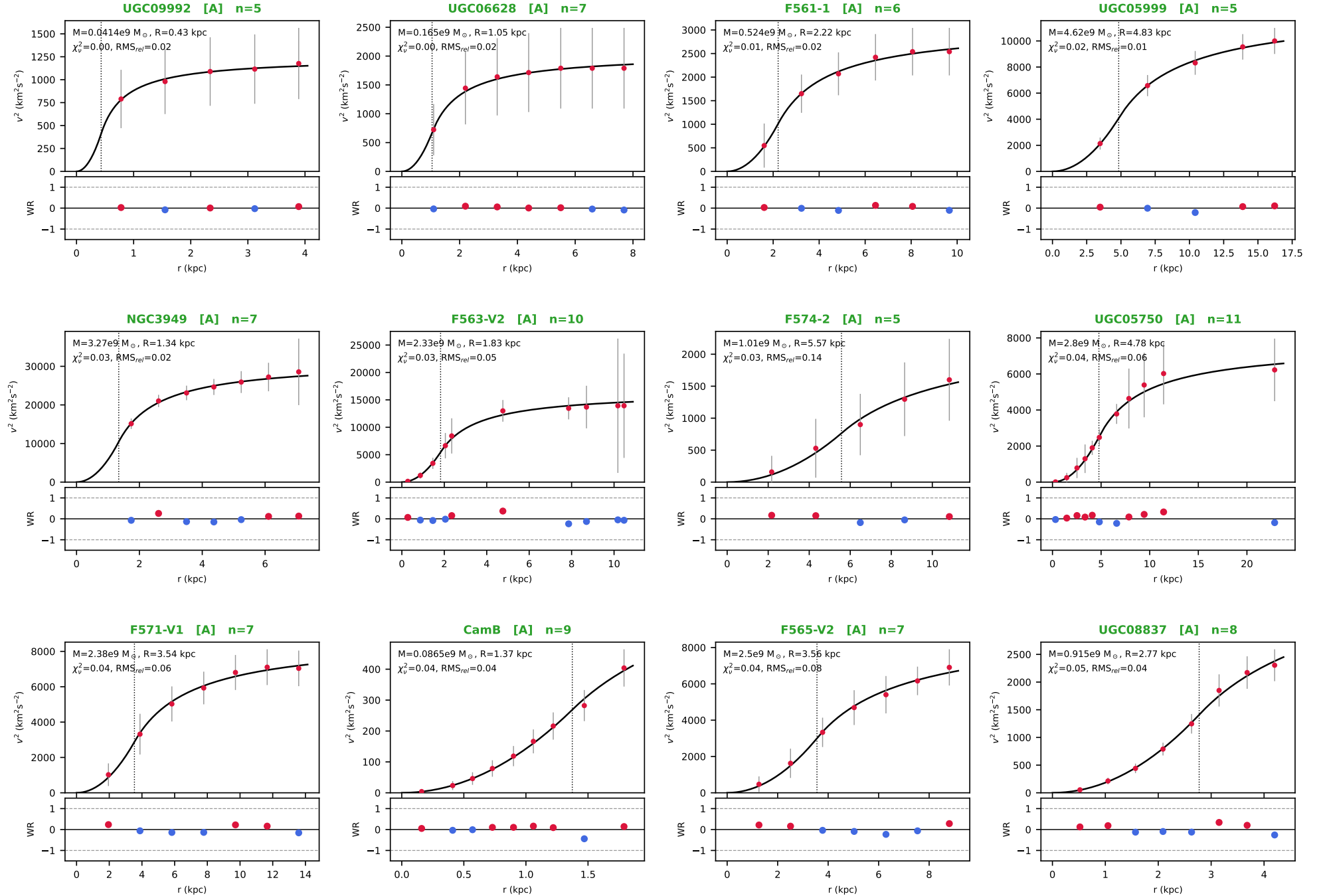
Table 4: Accepted two-L galaxies: goodness of fit. $AIC = \chi^2 + 2k$, $BIC = \chi^2 + k \ln n$; Δ relative to the best model of the galaxy; p_{runs} = runs test on the residuals of that model (small = residual structure remains).

Galaxy	Model	k	χ^2	χ^2_ν	AIC	BIC	ΔAIC	ΔBIC	RMS_{rel}	p_{runs}
NGC2403	Single-L	2	2274.86	32.040	2278.86	2283.44	1786.03	1779.15	0.251	3.6e-13
	Two-L	4	547.69	7.938	555.69	564.85	62.86	60.57	0.103	4.9e-09
	Two-L + Φ_{BH}^*	5	482.83	7.100	492.83	504.28	0.00	0.00	0.140	4.8e-11
UGC06787	Single-L	2	1792.40	25.977	1796.40	1800.92	1059.70	1052.91	0.247	1.7e-13
	Two-L	4	1160.82	17.326	1168.82	1177.87	432.12	429.86	0.143	9.2e-12
	Two-L + Φ_{BH}^*	5	726.70	11.011	736.70	748.02	0.00	0.00	0.137	3.3e-11
NGC5585	Single-L	2	636.55	28.934	640.55	642.91	547.60	544.06	0.400	1.1e-04
	Two-L	4	100.55	5.027	108.55	113.26	15.60	14.42	0.156	0.019
	Two-L + Φ_{BH}^*	5	82.95	4.366	92.95	98.84	0.00	0.00	0.124	0.002
IC2574	Single-L	2	420.08	13.128	424.08	427.14	386.54	381.96	0.251	6.8e-06
	Two-L	4	38.93	1.298	46.93	53.04	9.39	7.86	0.107	0.223
	Two-L + Φ_{BH}^*	5	27.55	0.950	37.55	45.18	0.00	0.00	0.086	0.348
UGC03580	Single-L	2	435.88	9.686	439.88	443.58	366.37	360.82	0.388	1.2e-10
	Two-L	4	141.27	3.285	149.27	156.67	75.76	73.91	0.337	4.5e-08
	Two-L + Φ_{BH}^*	5	63.51	1.512	73.51	82.76	0.00	0.00	0.087	3.1e-06
NGC0247	Single-L	2	205.93	8.580	209.93	212.45	198.41	194.63	0.243	8.7e-06
	Two-L	4	9.51	0.432	17.51	22.54	5.98	4.72	0.038	0.005
	Two-L + Φ_{BH}^*	5	1.53	0.073	11.53	17.82	0.00	0.00	0.028	0.008
NGC4013	Single-L	2	187.67	5.520	191.67	194.84	166.88	162.13	0.122	2.0e-07
	Two-L	4	86.85	2.714	94.85	101.18	70.06	68.48	0.123	2.0e-07
	Two-L + Φ_{BH}^*	5	14.79	0.477	24.79	32.71	0.00	0.00	0.046	1.0e-04
NGC1003	Single-L	2	167.05	4.913	171.05	174.21	129.33	124.58	0.148	1.0e-04
	Two-L	4	88.63	2.770	96.63	102.97	54.92	53.34	0.114	1.3e-05
	Two-L + Φ_{BH}^*	5	31.71	1.023	41.71	49.63	0.00	0.00	0.125	0.002
NGC7793	Single-L	2	113.06	2.569	117.06	120.72	71.11	67.46	0.303	4.4e-08
	Two-L*	4	37.94	0.903	45.94	53.26	0.00	0.00	0.213	0.002
	Two-L + Φ_{BH}	5	37.56	0.916	47.56	56.71	1.62	3.45	0.131	0.001
NGC3741	Single-L	2	66.79	3.515	70.79	72.88	56.13	54.04	0.365	7.5e-05
	Two-L*	4	6.66	0.392	14.66	18.83	0.00	0.00	0.134	0.148
	Two-L + Φ_{BH}	5	4.75	0.297	14.75	19.97	0.09	1.13	0.069	0.148
DDO161	Single-L	2	62.32	2.149	66.32	69.19	47.93	45.07	0.348	9.0e-05
	Two-L*	4	10.39	0.385	18.39	24.13	0.00	0.00	0.096	0.001
	Two-L + Φ_{BH}	5	8.41	0.323	18.41	25.58	0.02	1.45	0.039	0.062
DDO154	Single-L	2	74.33	7.433	78.33	79.30	46.62	45.16	0.210	0.038
	Two-L	4	27.80	3.475	35.80	37.74	4.09	3.60	0.141	0.272
	Two-L + Φ_{BH}^*	5	21.71	3.102	31.71	34.14	0.00	0.00	0.067	0.541
F571-8	Single-L	2	41.73	3.794	45.73	46.86	33.78	32.65	0.347	0.005
	Two-L*	4	3.96	0.440	11.96	14.22	0.00	0.00	0.132	0.239
	Two-L + Φ_{BH}	5	3.67	0.459	13.67	16.49	1.71	2.28	0.102	0.394
UGC04278	Single-L	2	45.40	1.974	49.40	51.84	34.08	31.64	0.381	1.4e-05
	Two-L*	4	7.32	0.349	15.32	20.20	0.00	0.00	0.141	3.2e-04
	Two-L + Φ_{BH}	5	7.24	0.362	17.24	23.33	1.91	3.13	0.128	0.001
UGC12732	Single-L	2	39.35	2.811	43.35	44.90	30.74	28.42	0.209	9.9e-04
	Two-L	4	10.77	0.898	18.77	21.86	6.16	5.39	0.127	0.021
	Two-L + Φ_{BH}^*	5	2.61	0.238	12.61	16.48	0.00	0.00	0.054	0.065
UGC02885	Single-L	2	34.46	2.027	38.46	40.35	19.18	17.29	0.098	0.005
	Two-L*	4	11.28	0.752	19.28	23.06	0.00	0.00	0.056	0.017
	Two-L + Φ_{BH}	5	16.48	1.177	26.48	31.21	7.21	8.15	0.065	0.019
UGC00731	Single-L	2	23.40	2.340	27.40	28.37	18.08	17.11	0.175	0.008
	Two-L*	4	1.32	0.165	9.32	11.26	0.00	0.00	0.030	0.272
	Two-L + Φ_{BH}	5	1.17	0.168	11.17	13.60	1.85	2.34	0.028	0.541
UGC07524	Single-L	2	31.09	1.072	35.09	37.96	20.75	16.60	0.249	6.7e-05
	Two-L*	4	7.63	0.282	15.63	21.36	1.28	0.00	0.127	0.010
	Two-L + Φ_{BH}	5	4.34	0.167	14.34	21.51	0.00	0.15	0.078	0.035

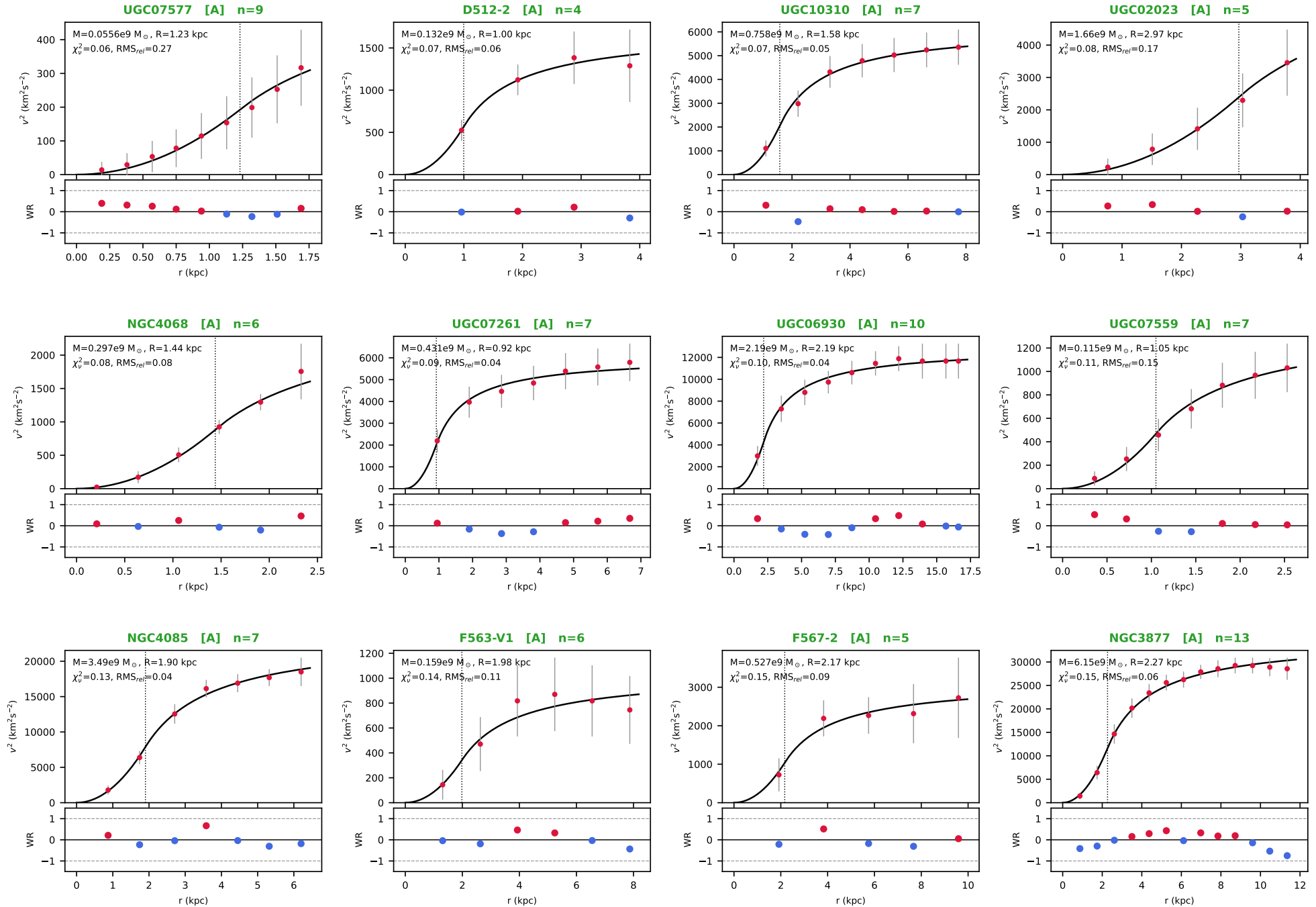
Galaxy	Model	k	χ^2	χ^2_ν	AIC	BIC	ΔAIC	ΔBIC	RMS_{rel}	p_{runs}
ESO116-G012	Single-L	2	33.41	2.570	37.41	38.83	19.75	17.63	0.259	0.007
	Two-L	4	16.34	1.485	24.34	27.17	6.68	5.97	0.194	0.092
	Two-L + Φ_{BH}^*	5	7.66	0.766	17.66	21.20	0.00	0.00	0.079	0.215
D631-7	Single-L	2	18.68	1.335	22.68	24.23	11.99	10.45	0.184	0.002
	Two-L*	4	2.69	0.224	10.69	13.78	0.00	0.00	0.049	0.150
	Two-L + Φ_{BH}	5	2.69	0.245	12.69	16.55	2.00	2.77	0.049	0.150
UGC06614	Single-L	2	17.86	1.624	21.86	22.99	10.93	9.80	0.158	0.005
	Two-L*	4	2.93	0.325	10.93	13.19	0.00	0.00	0.054	0.007
	Two-L + Φ_{BH}	5	1.75	0.219	11.75	14.58	0.82	1.39	0.062	0.197
NGC0055	Single-L	2	19.78	1.041	23.78	25.87	10.20	8.11	0.135	1.3e-04
	Two-L*	4	5.58	0.328	13.58	17.76	0.00	0.00	0.056	0.007
	Two-L + Φ_{BH}	5	3.70	0.231	13.70	18.92	0.12	1.16	0.045	0.025
NGC3726	Single-L	2	29.76	2.976	33.76	34.73	10.21	8.76	0.099	0.038
	Two-L*	4	17.50	2.187	25.50	27.44	1.95	1.46	0.091	0.409
	Two-L + Φ_{BH}	5	13.55	1.936	23.55	25.98	0.00	0.00	0.069	0.338

Atlas 1: single-L fits with residuals (all 175, grouped by category)

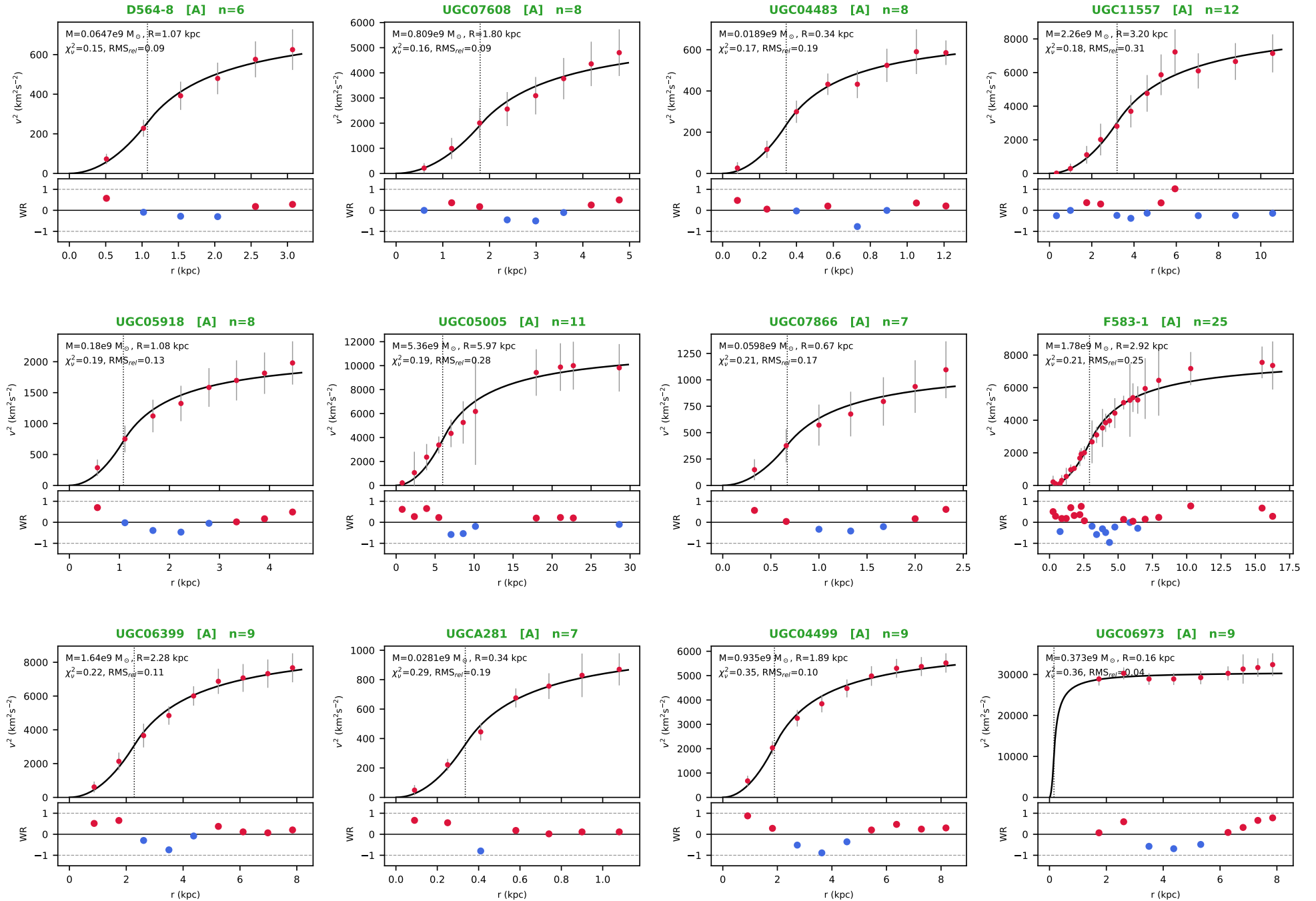
Step 1-2: single-L CL fits and weighted residuals $WR=(v_{obs}^2-v_{mod}^2)/\sigma_{v^2}$ [A adequate, B ++ wave, C +- wave, D structured other, E poor unstructured]; orange line: first crossover



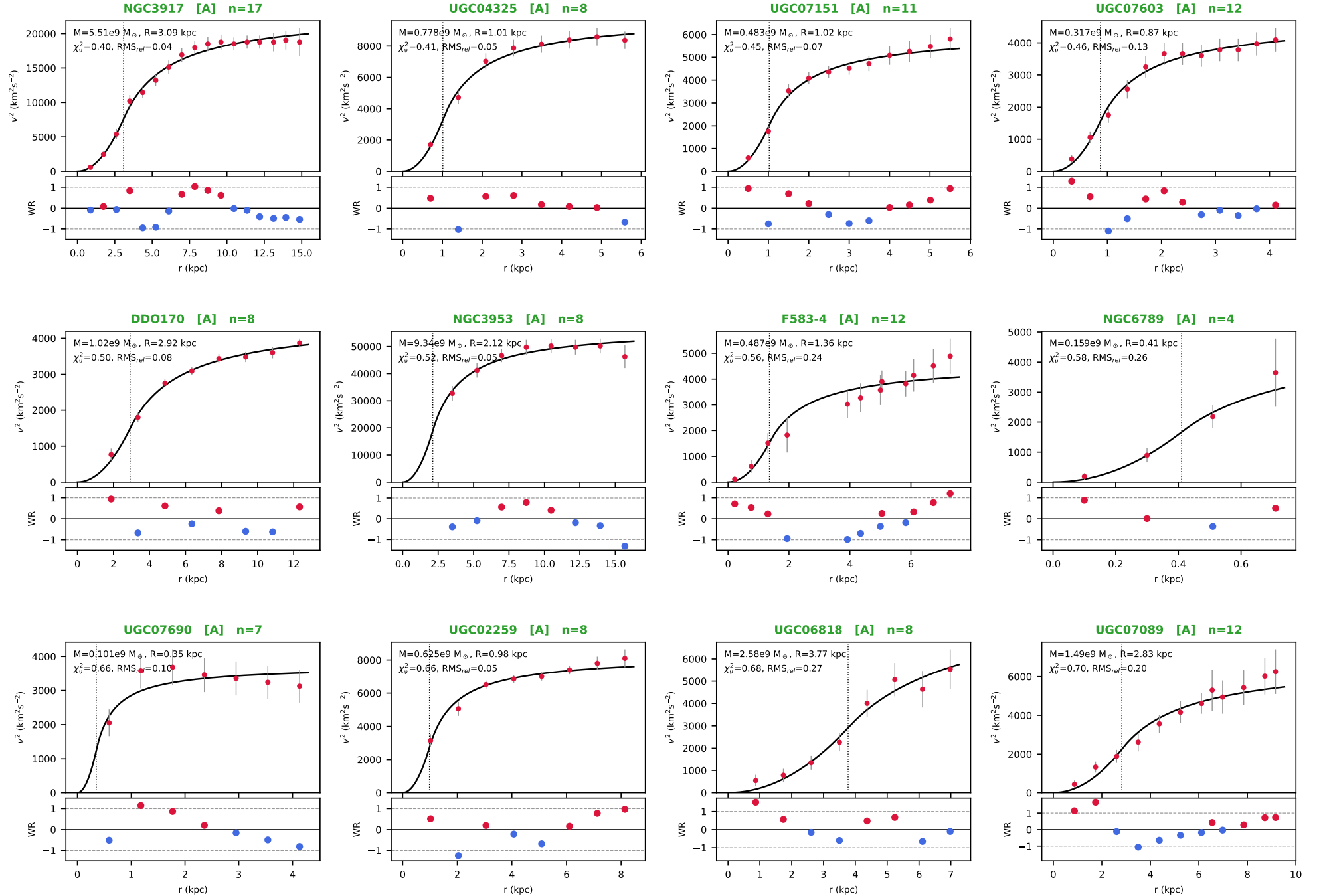
Step 1-2: single-L CL fits and weighted residuals $WR=(v_{obs}^2-v_{mod}^2)/\sigma_{v^2}$ [A adequate, B +-+ wave, C +- wave, D structured other, E poor unstructured]; orange line: first crossover



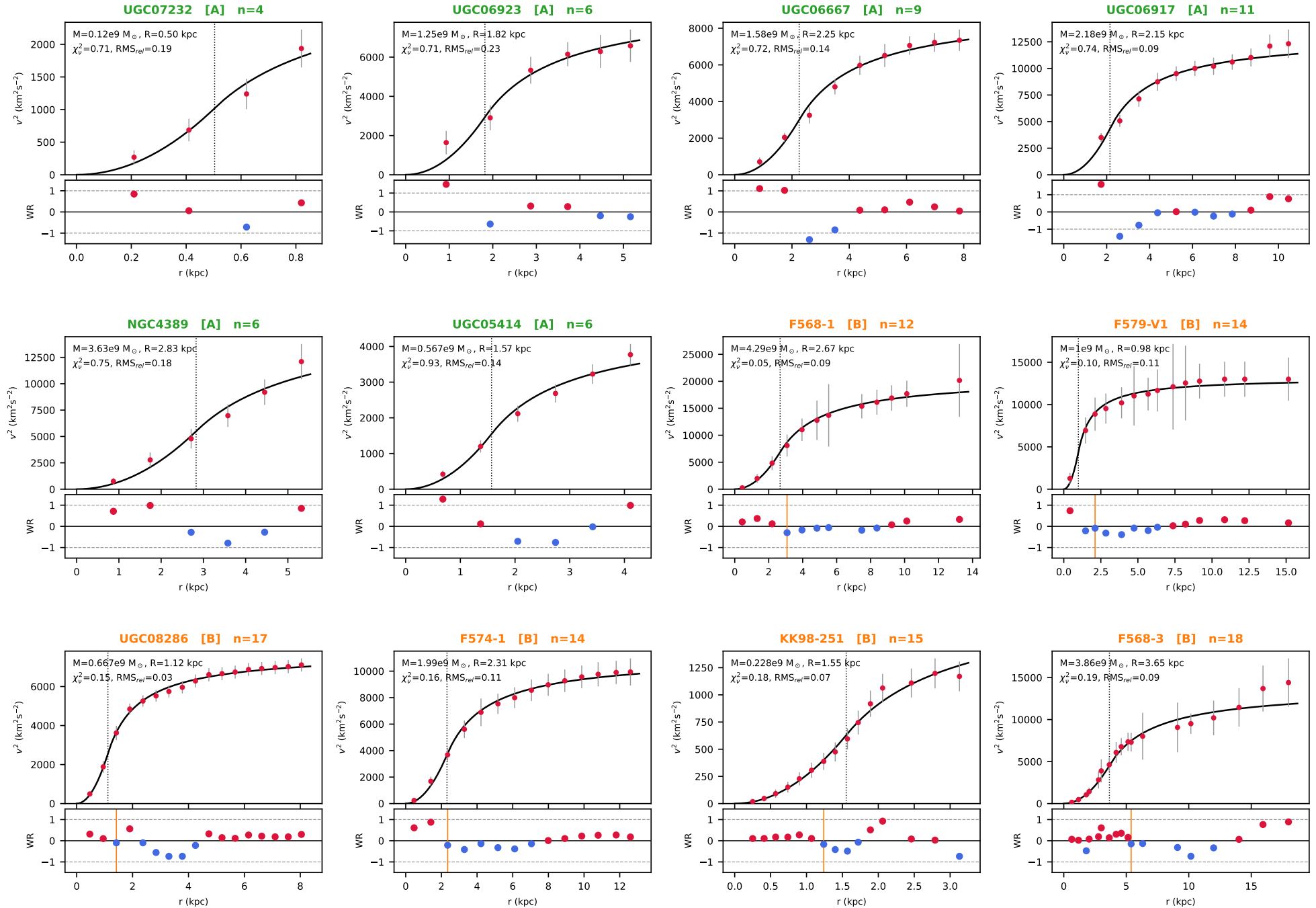
Step 1-2: single-L CL fits and weighted residuals $WR=(v_{obs}^2-v_{mod}^2)/\sigma_{v^2}$ [A adequate, B +- wave, C +- wave, D structured other, E poor unstructured]; orange line: first crossover



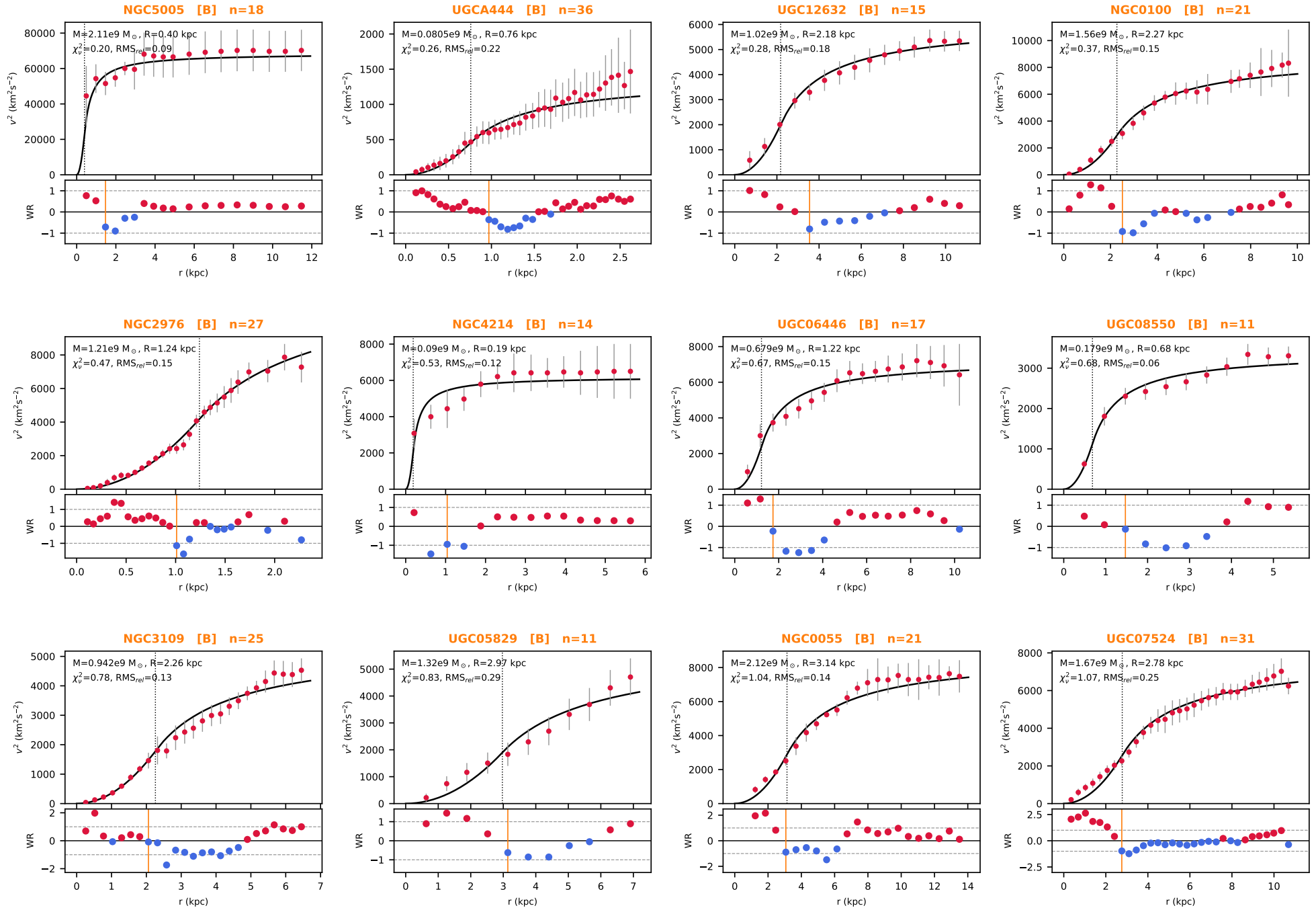
Step 1-2: single-L CL fits and weighted residuals $WR=(v_{obs}^2-v_{mod}^2)/\sigma_{v^2}$ [A adequate, B +++ wave, C +- wave, D structured other, E poor unstructured]; orange line: first crossover



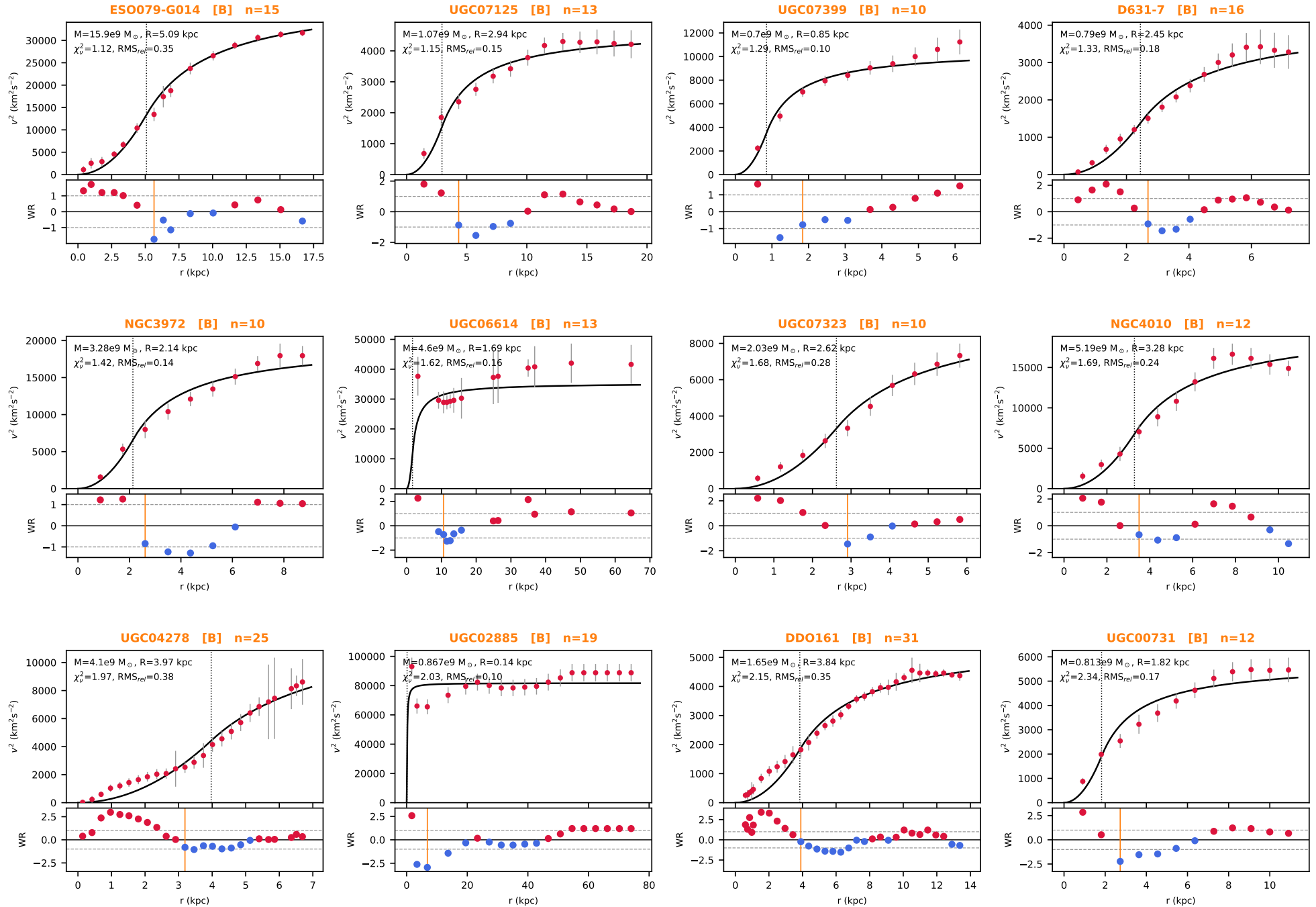
Step 1-2: single-L CL fits and weighted residuals $WR=(v_{obs}^2-v_{mod}^2)/\sigma_{v^2}$ [A adequate, B ++ wave, C +- wave, D structured other, E poor unstructured]; orange line: first crossover



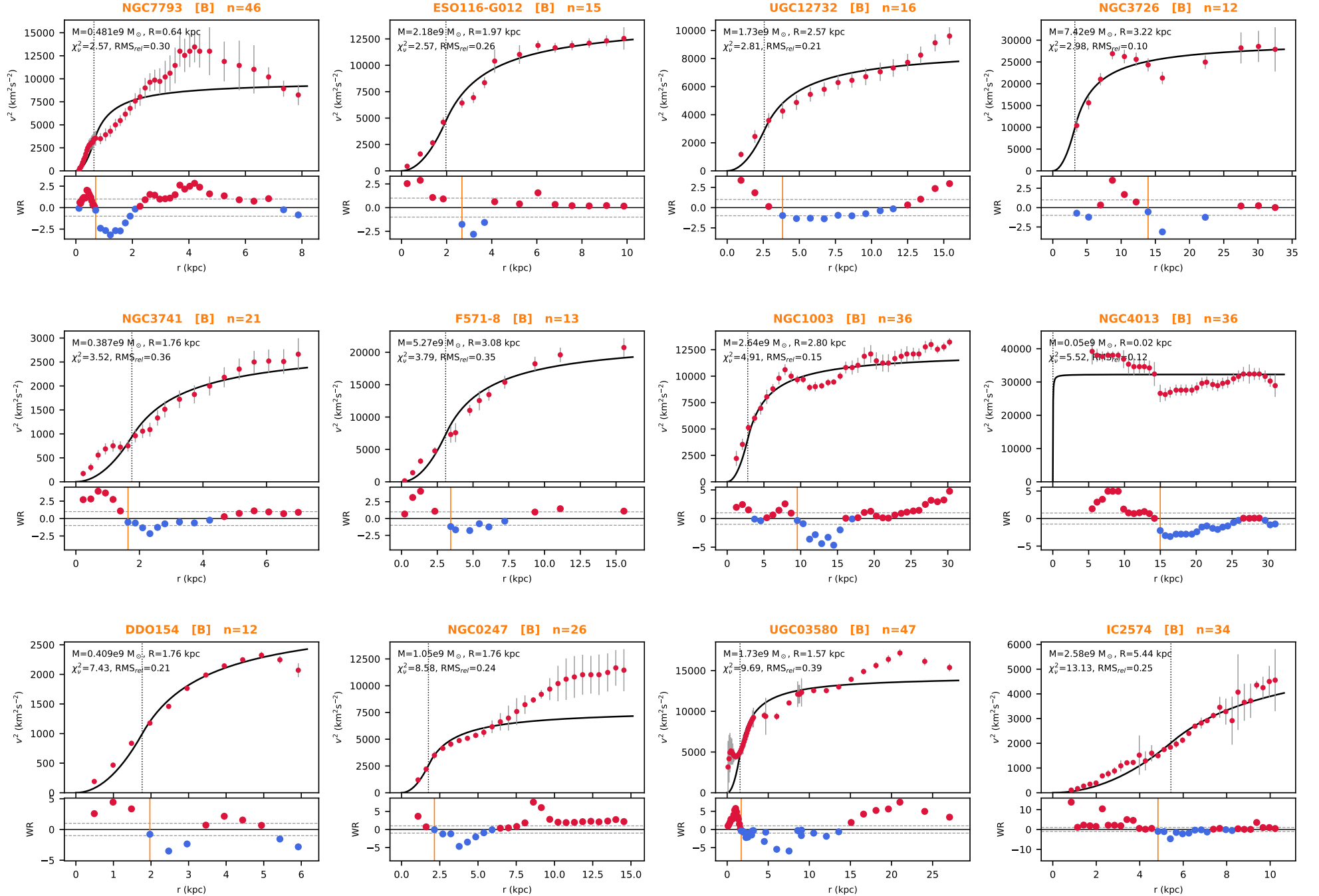
Step 1-2: single-L CL fits and weighted residuals $WR=(v_{obs}^2-v_{mod}^2)/\sigma_{v^2}$ [A adequate, B ++ wave, C +- wave, D structured other, E poor unstructured]; orange line: first crossover



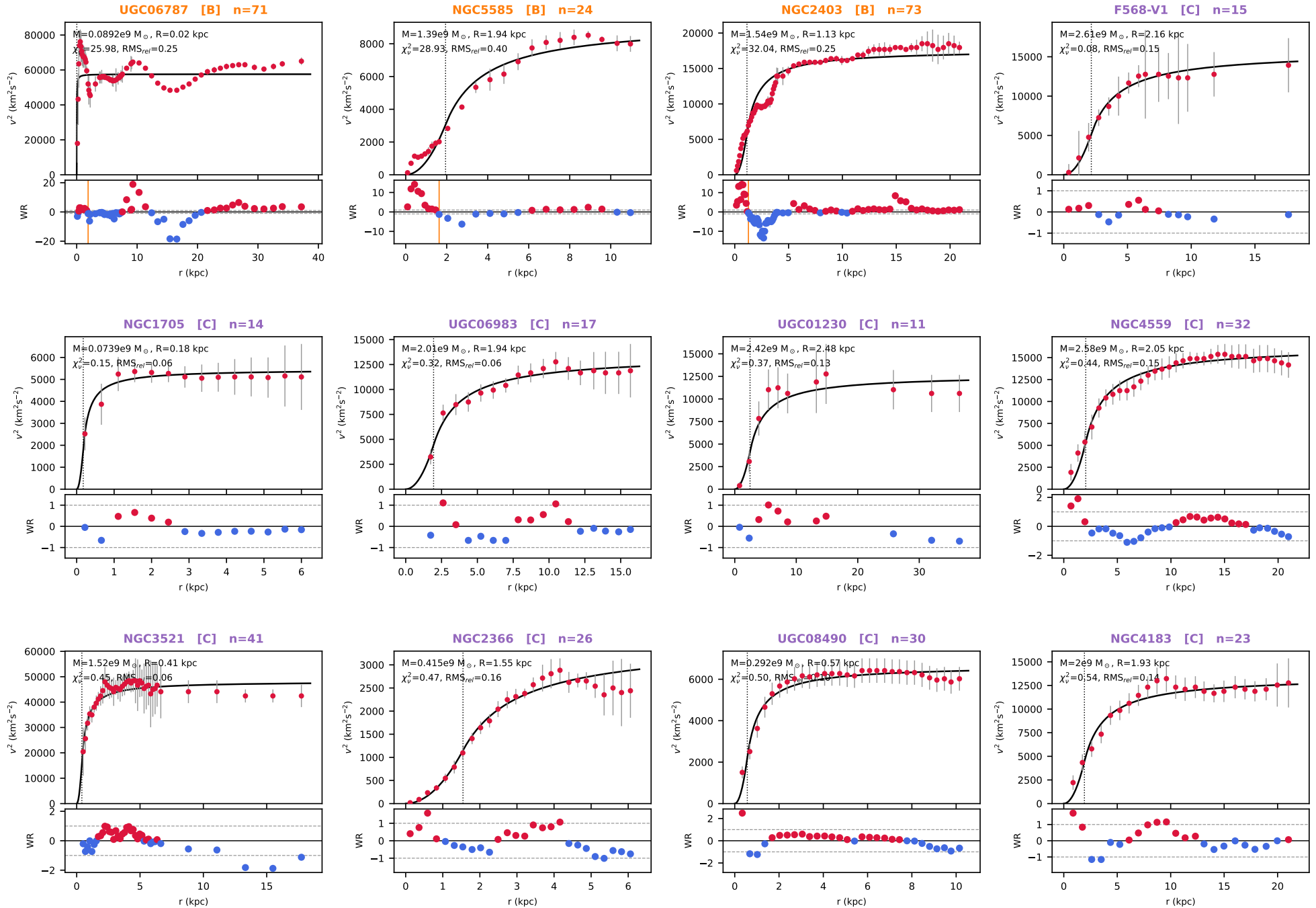
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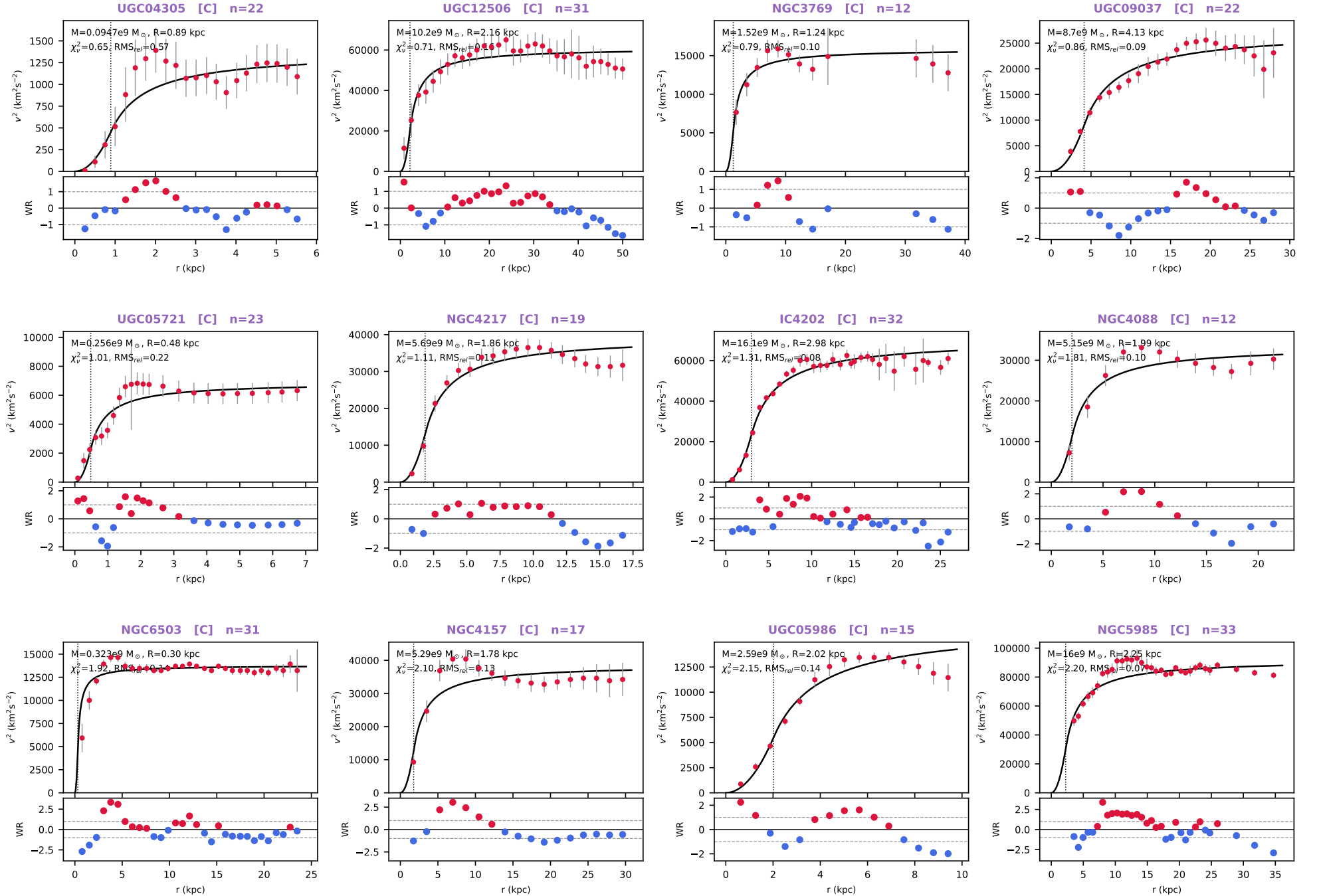
Step 1-2: single-L CL fits and weighted residuals $WR=(v_{obs}^2-v_{mod}^2)/\sigma_{v^2}$ [A adequate, B ++ wave, C +- wave, D structured other, E poor unstructured]; orange line: first crossover



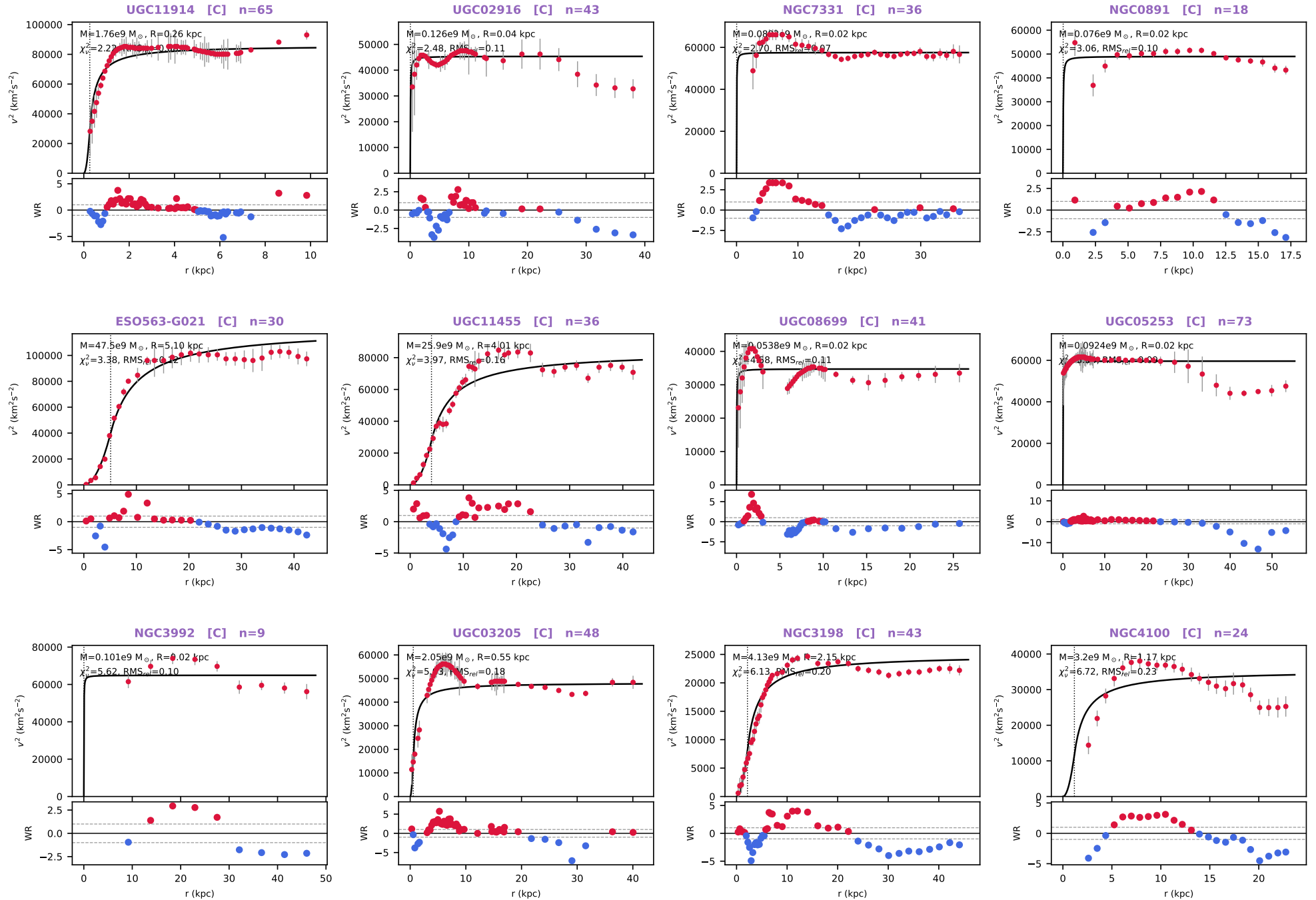
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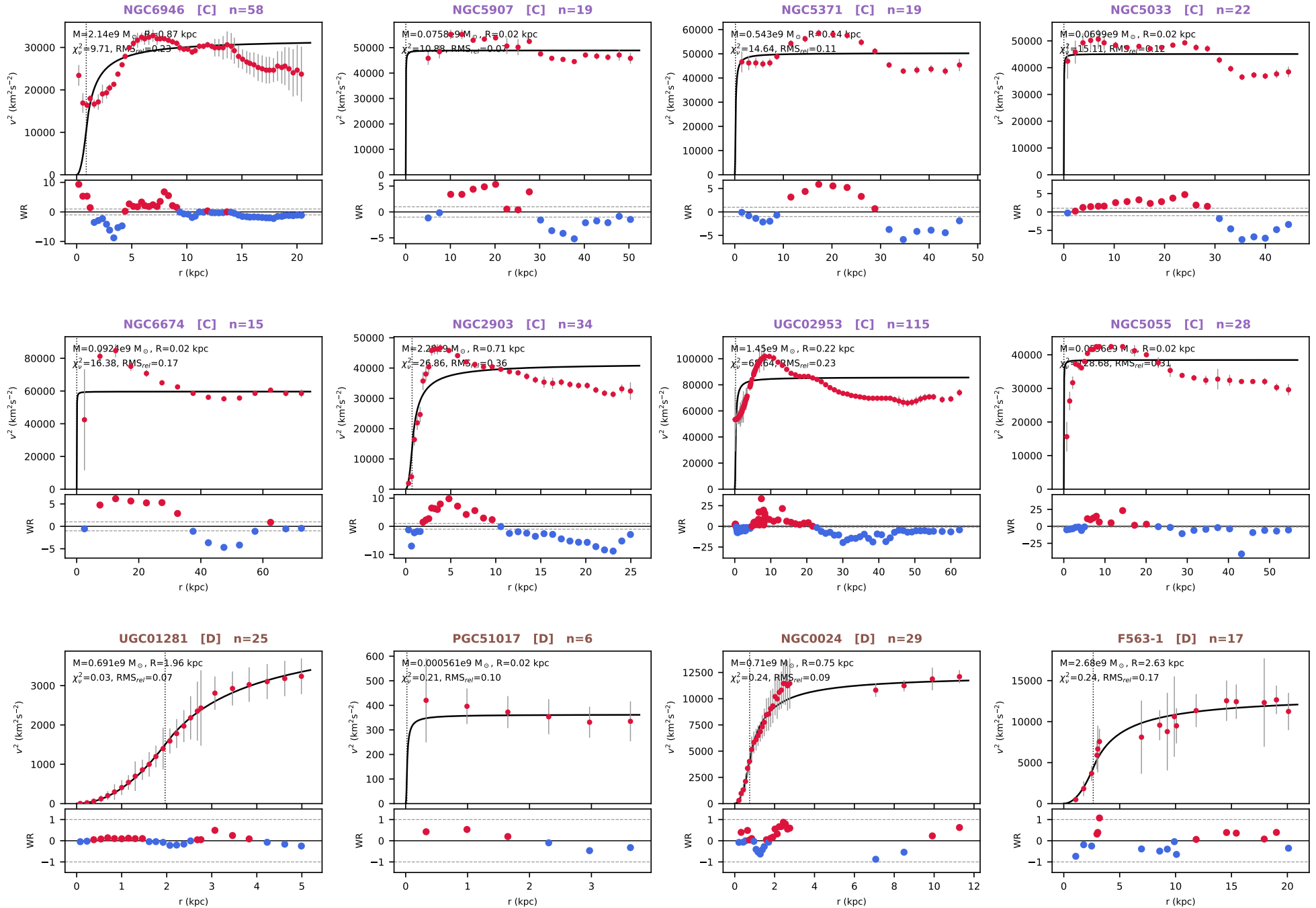
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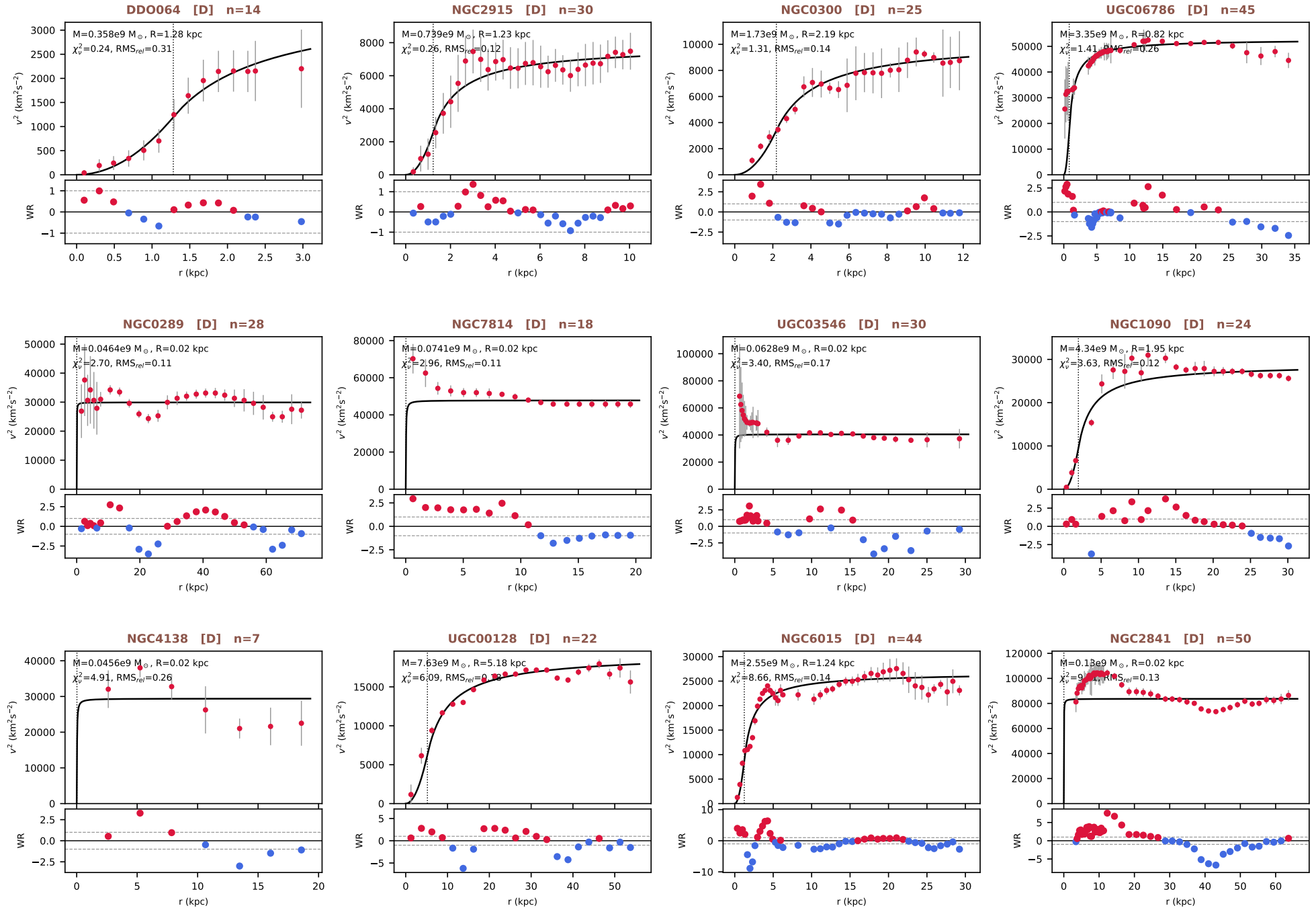
Step 1-2: single-L CL fits and weighted residuals $WR=(v_{obs}^2-v_{mod}^2)/\sigma_{v^2}$ [A adequate, B ++ wave, C -+- wave, D structured other, E poor unstructured]; orange line: first crossover



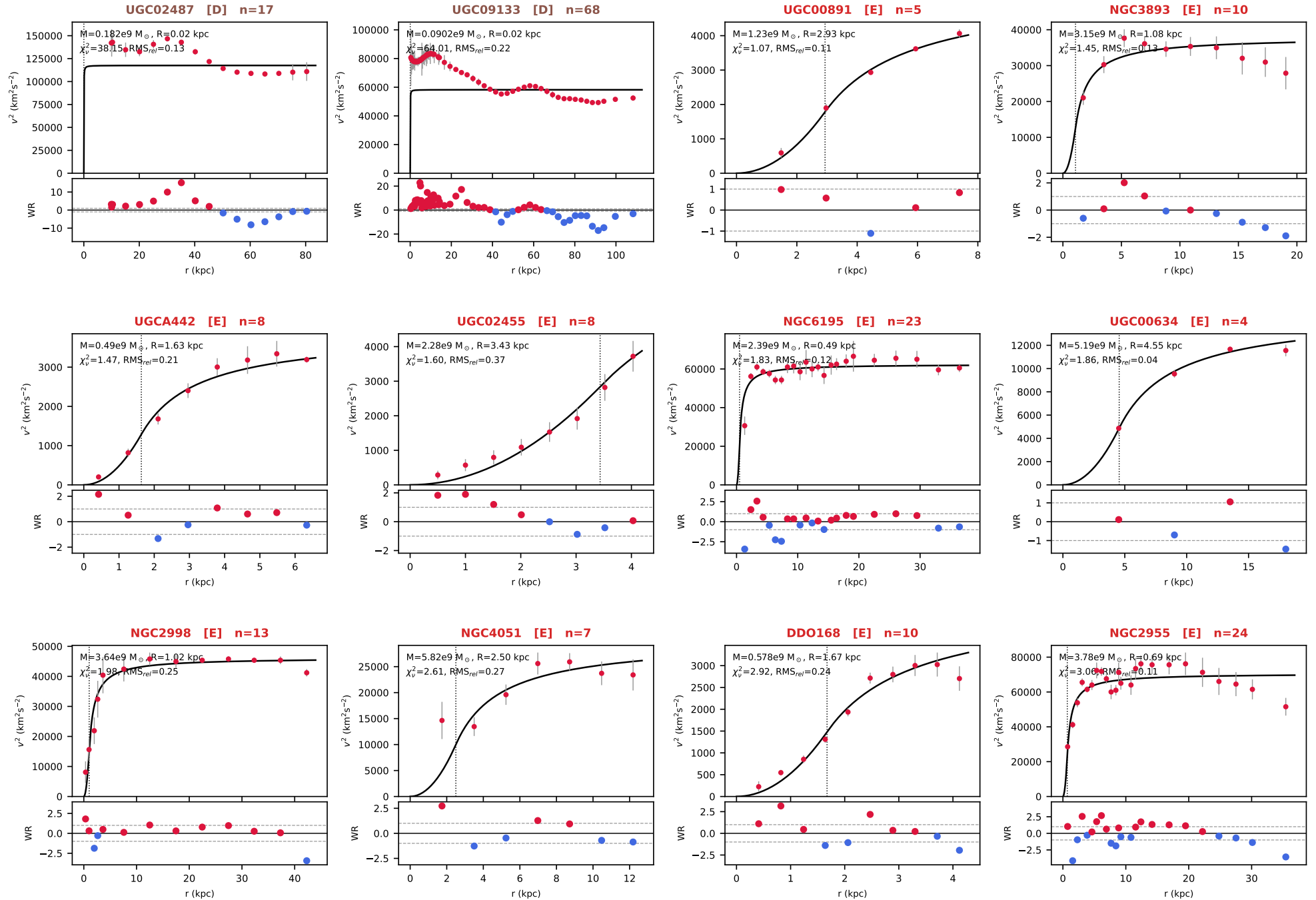
Step 1-2: single-L CL fits and weighted residuals $WR=(v_{obs}^2-v_{mod}^2)/\sigma_{v^2}$ [A adequate, B ++ wave, C +- wave, D structured other, E poor unstructured]; orange line: first crossover



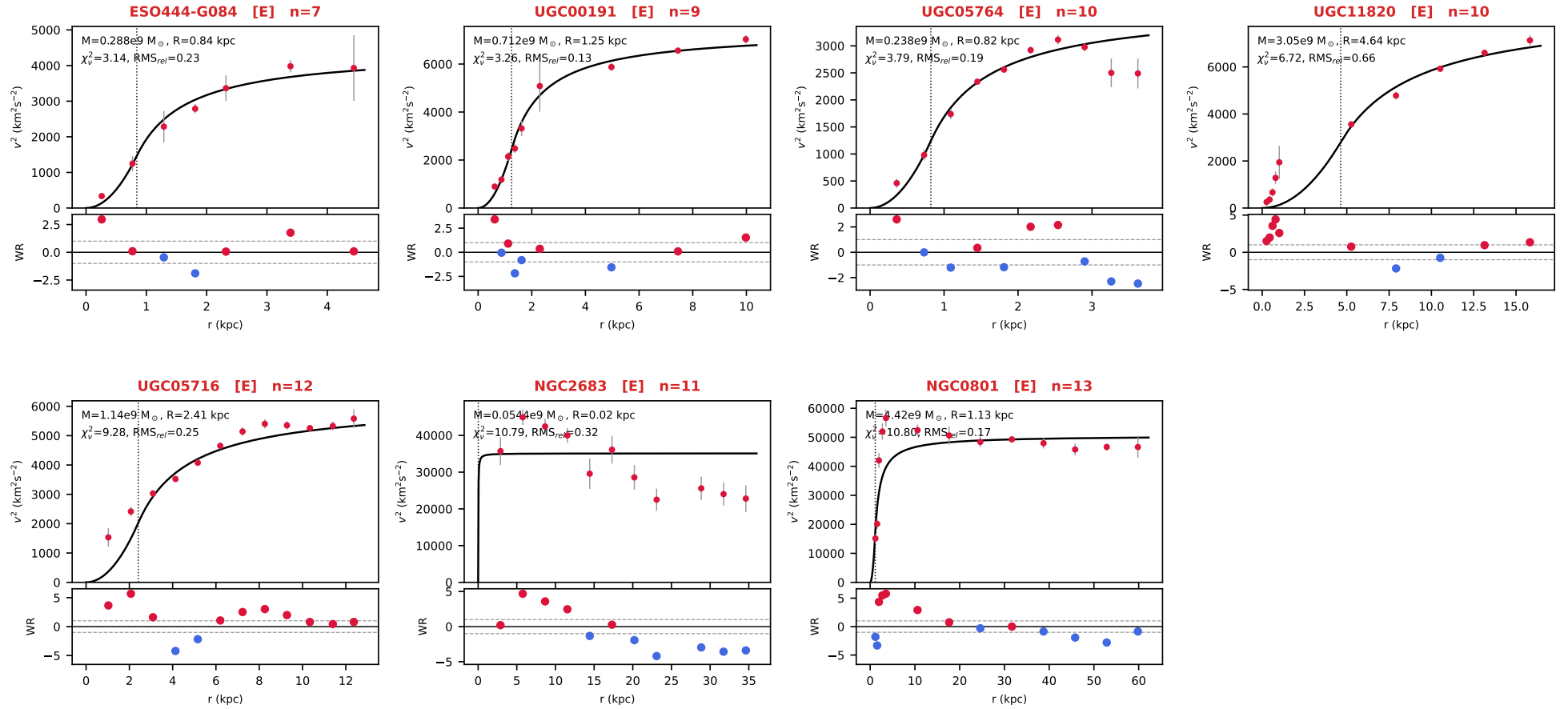
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Step 1-2: single-L CL fits and weighted residuals $WR=(v_{obs}^2-v_{mod}^2)/\sigma_{v^2}$ [A adequate, B +-+ wave, C -+- wave, D structured other, E poor unstructured]; orange line: first crossover

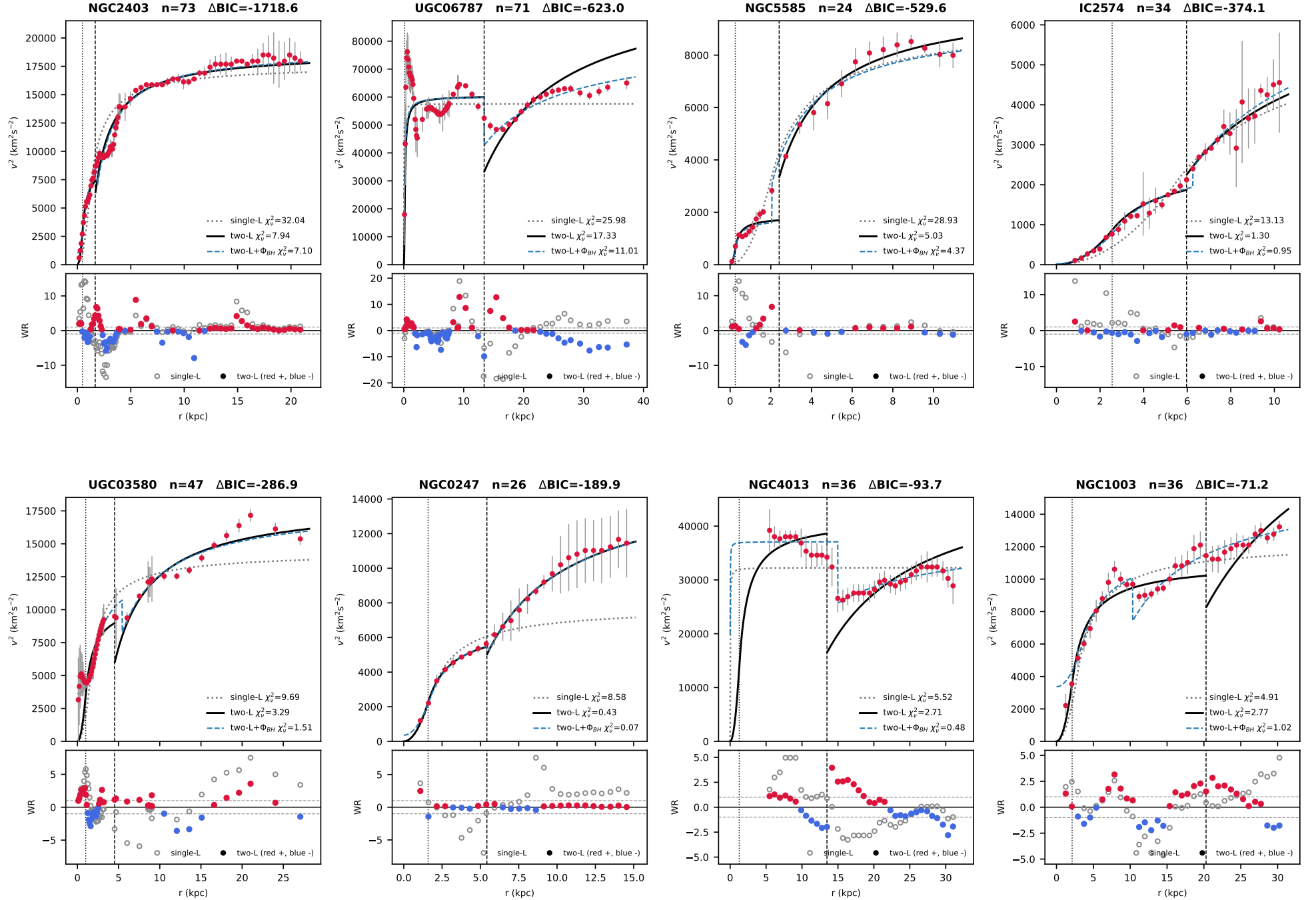


Step 1-2: single-L CL fits and weighted residuals $WR=(v_{obs}^2-v_{mod}^2)/\sigma_{v^2}$ [A adequate, B ++ wave, C +- wave, D structured other, E poor unstructured]; orange line: first crossover

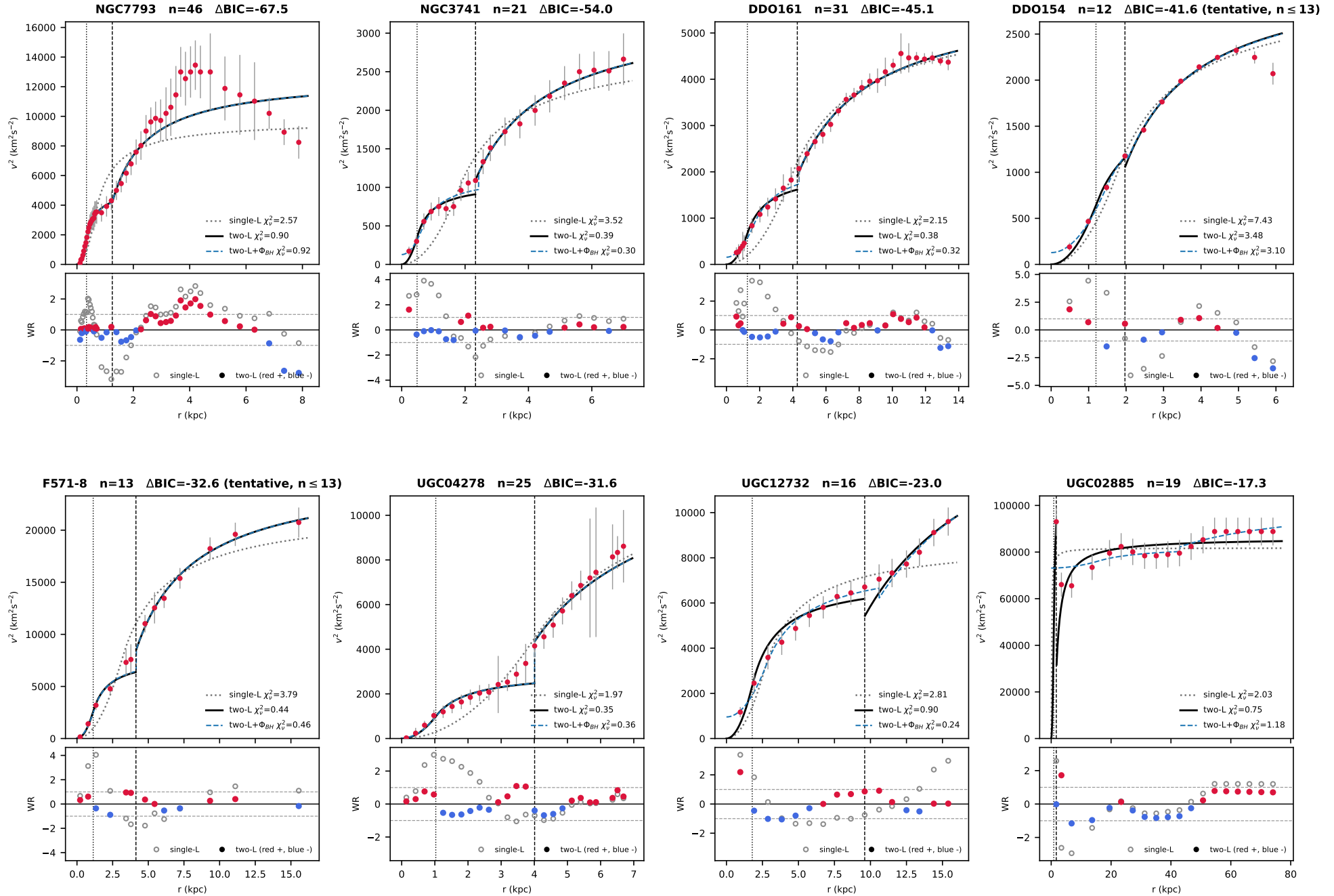


Atlas 2: accepted two-L fits with residuals

Steps 3-5: accepted two-Lagrangian fits (Eqs. 24-26), R_1 dotted, R_2 dashed; lower panels: weighted residuals of single-L (open grey) and two-L (filled)



Steps 3-5: accepted two-Lagrangian fits (Eqs. 24-26), R_1 dotted, R_2 dashed; lower panels: weighted residuals of single-L (open grey) and two-L (filled)



Steps 3-5: accepted two-Lagrangian fits (Eqs. 24-26), R_1 dotted, R_2 dashed; lower panels: weighted residuals of single-L (open grey) and two-L (filled)

